

The GreatWall Signature Scheme

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Tuesday 30th June, 2026

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1 Introduction

This document specifies **GreatWall**, a family of post-quantum digital signature schemes based solely on one-way function assumptions. GreatWall follows the *VOLE-in-the-Head* (VOLEitH) paradigm [9] and instantiates the public-key relation with a dedicated symmetric one-way function, called Pylon. The scheme is designed to provide existential unforgeability under adaptive chosen-message attacks while avoiding number-theoretic, lattice-based, code-based, or multivariate public-key assumptions. And it consists of four variants—GreatWall-128, GreatWall-192, GreatWall-256, and GreatWall-512—providing different levels of classical and post-quantum security, respectively.

The motivation behind GreatWall is the growing need for post-quantum signatures with compact public keys, predictable performance, and a scalable security path. Existing symmetric-key-based signature schemes have demonstrated that VOLEit techniques can yield practical post-quantum signatures, but their efficiency and size are strongly influenced by the cost of proving the underlying one-way-function relation. For example, the size of the extended witness and the number of nonlinear constraints directly affect signature size, signing time, and verification time. GreatWall aims at the trade-off between efficiency and security through a co-design of the signature layer and the underlying one-way function: the proof system is kept within the VOLEitH framework, while Pylon is constructed to minimize proof cost without introducing an algebraically fragile relation.

At a high level, a GreatWall signature is a non-interactive zero-knowledge proof of knowledge of a preimage of the public key under Pylon. The proof is obtained by applying the Fiat–Shamir transform [20] to a VOLE-based proof system. The signer commits to VOLE-related data using a GGM-tree-based batch all-but-one vector commitment, derives challenges through random oracles, and proves consistency of the Pylon relation through QuickSilver-style multiplication checks. Since linear operations are handled locally by the linearity of VOLE correlations, the dominant proof cost comes from the nonlinear part of Pylon. The security of the resulting scheme reduces entirely to the one-wayness of the underlying symmetric primitive and the soundness of the VOLEitH proof system, with no structured algebraic hardness assumption introduced. All the variants share the same construction; only the parameters of the one-way function and the proof system are adjusted across security levels, so that a single design instruments the entire family.

The dedicated one-way function, dubbed **Pylon**, is the core of the construction. Pylon is a symmetric primitive defined over a binary extension field, designed jointly with the VOLEitH proof system in order to keep the witness encoding compact and the consistency check structurally simple. Its central design feature is a *nested feed-forward* structure: an outer feed-forward branch spans the entire function and binds the secret input to the global input-output relation, while two inner feed-forward branches act on local sub-structures within the outer one. This nested arrangement is the result of a deliberate trade-off. The inner branches enrich the algebraic complexity of each round at almost no

additional multiplicative cost, while the outer branch ties the local algebraic obstructions together into a single global one-wayness relation. The construction therefore retains a small number of nonlinear constraints per round, which is essential for an efficient VOLEitH proof, while substantially raising the cost of any algebraic attack that attempts to peel off rounds independently. The detailed specifications and recommended parameters of Pylon are given in Sect. 2.2.

The security analysis of GreatWall follows a two-level approach. At the signature level, EUF-CMA security follows from the soundness and zero-knowledge properties of the VOLEitH proof system, together with the Fiat–Shamir transform in the random-oracle model and its post-quantum analogue in the quantum random-oracle model. At the primitive level, the required assumption is the preimage resistance of Pylon. This document supports the Pylon assumption through both a generic black-box analysis in the ideal-permutation model and dedicated cryptanalytic evaluations, including brute-force search, Gröbner-basis attacks, overdefined quadratic modeling, resultant attacks, subgroup-linearization attacks, statistical attacks, and so on.

GreatWall has been designed for robustness and scalability. It provides four parameter variants, namely GreatWall-128, GreatWall-192, GreatWall-256, and GreatWall-512. These variants share the same overall architecture and differ only in field size, exponent choices, and VOLEitH parameters. This uniform structure simplifies implementation and analysis, while allowing the scheme to scale naturally to higher security levels. Within the VOLEitH framework, this scalability translates directly into predictable signature sizes and predictable signing and verification cost across the entire family. The performance of GreatWall has been benchmarked against state-of-the-art post-quantum signature schemes built on symmetric primitives, with particular attention to leading VOLEitH-based designs. As a result, GreatWall is particularly well suited to applications that require long-term signature validity together with moderate signing latency, including firmware updates, code signing, and archival authentication.

The construction deliberately avoids assumptions vulnerable to Shor-type [33] quantum algorithms. The main generic quantum speedup relevant to Pylon is Grover search, which is accounted for in the parameter selection. Together with its compact-key structure, scalable parameterization, and proof-cost-aware one-way-function design, GreatWall is intended to be a practical candidate for long-term post-quantum authentication scenarios such as firmware signing, code signing, secure boot, software update distribution, and archival authentication.

2 Specification

This section presents a complete description of the GreatWall family, starting with the basic notations and definitions in Sect. 2.1, the specification of the one-way function Pylon in Sect. 2.2 and then an overview of the algorithms in Sect. 2.3.

Table 1: Notations

Notation	Description
$[n]$	The set $\{1, \dots, n\}$ for any positive integer n .
$\text{hw}(n)$	The Hamming weight of n , i.e., the number of non-zero bits in its binary representation.
$x \leftarrow a$	Assignment of the value a to the variable x .
$y \xleftarrow{\$} S$	Sampling y uniformly at random from the set S .
$+$	Addition in the corresponding algebraic structure, including extension fields and integers.
$\oplus$	Bitwise exclusive-OR operation.
$x^{(i)}$	The share of a secret x held by the i -th party in additive secret sharing, where the sum of all shares equals x .
λ	security parameter in classical settings.
n	Input or output bit length of S-boxes.
r	Total number of rounds of Pylon.
τ	Number of parallel repetitions required to achieve the target security level in the Non-Interactive Zero-Knowledge Proof of Knowledge (NIZKPoK) protocol.
N	Total number of parties in the proof system.
$\text{negl}(\lambda)$	A negligible function in λ if it vanishes faster than the reciprocal of any polynomial in λ .
PPT	Probabilistic polynomial-time.

2.1 Notations and Definitions

Digital Signatures. A digital signature scheme consists of three PPT algorithms ($\text{Gen}, \text{Sign}, \text{Verify}$):

- Key generation: $\text{Gen}(1^\lambda)$ takes a security parameter λ as input and outputs a pair of public key and secret key (pk, sk) .
- Signing: $\text{Sign}(sk, M)$ produces a signature σ of the message $M \in \{0, 1\}^*$ with the secret key sk .
- Verification: $\text{Verify}(pk, M, \sigma)$ will return 1 if σ is valid for M and sk , and 0 otherwise.

A valid signature scheme should satisfy that for all honestly generated key pairs and all messages M , $\text{Verify}(pk, M, \text{Sign}(sk, M)) = 1$. Informally, a signature scheme can be considered as secure if no efficient adversary, after adaptively querying a signing oracle on chosen messages, can produce a valid signature on a new message with non-negligible probability. And Definition 1 captures this.

Definition 1. *Given a signature $\Sigma = (\text{Gen}, \text{Sign}, \text{Verify})$, if for any PPT adversary $\mathcal{A}$, there always exists a negligible function $\text{negl}(\lambda)$ such that*

$$\begin{aligned}
 & \Pr \left[(pk, sk) \leftarrow \text{Gen}(1^\lambda), \right. \\
 & \quad (M', \sigma') \leftarrow \mathcal{A}^{\text{Sign}(sk, \cdot)}(pk, 1^\lambda), \\
 & \quad \left. M' \notin Q, \text{Verify}(pk, M', \sigma') = 1 \right] < \text{negl}(\lambda).
 \end{aligned} \tag{1}$$

where $\mathcal{A}^{\text{Sign}(sk, \cdot)}(pk, 1^\lambda)$ denotes that $\mathcal{A}$ has access to the oracle $\text{Sign}(sk, \cdot)$ with (pk, sk) already generated, Q is the set of the queries made by $\mathcal{A}$. Then we can say that the signature Σ is secure against existential unforgeability under chosen message attack (EUF-CMA), or EUF-CMA secure.

In the post-quantum setting, the adversary $\mathcal{A}$ would be modeled as a quantum polynomial-time algorithm, while the accessible signing oracle remains classical. The existence of secure digital signature schemes is fundamentally tied to the existence of one-way functions (OWFs). Specifically, it has been theoretically proven by Rompel that OWFs are both a necessary and sufficient condition for constructing secure signatures [32].

One-Way functions. A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is an OWF only when two conditions can be satisfied, $f(x)$ is polynomial-time computable for a randomly chosen input $x \xleftarrow{\$} \{0, 1\}^*$, and computationally infeasible for any polynomial-time adversary to recover x with non-negligible probability. OWFs can be instantiated from symmetric cryptographic primitives such as block ciphers, permutations, or hash-function-based constructions. In contrast to number-theoretic assumptions, the security of symmetric primitives in the presence of quantum adversaries is comparatively well understood. Known quantum algorithms, most notably Grover’s algorithm [24], provide at most a quadratic speedup for generic attacks.

2.2 Specification of Pylon

A new one-way function, Pylon, serves as the core building block of GreatWall. The name reflects its role as a simple yet load-bearing component supporting the overall VOLEitH signature scheme. This new one-way function is constructed based on an *iterative, nested feed-forward structure*, and consists of a nonlinear operation S and a constant addition over large field $\mathbb{F}_{2^n}$, together with a linear layer.

Given input/output size n and exponents $(e_1, e_2) \in \mathbb{Z}^2$, Pylon: $\{0, 1\}^n \times \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is defined as

$$\begin{cases} x_1 = S_1(M_1(pt) + \gamma_1), \\ x_2 = S_2(M_2(x_1 + pt) + \text{iv}_{\text{owf}}), \\ ct = M_3(x_2) + M_2(x_1 + pt) + \text{iv}_{\text{owf}} + pt + \gamma_2. \end{cases} \quad (2)$$

And the overview of our construction is captured in Figure 1.

Nonlinear Components.

The nonlinear layer $S : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$, denoted as $\text{Mer}[e]^{-1}$, is instantiated as an inverse Mersenne mapping,

$$S(x) = x^{\bar{e}}, \quad \bar{e} = (2^e - 1)^{-1} \bmod (2^n - 1), \quad (3)$$

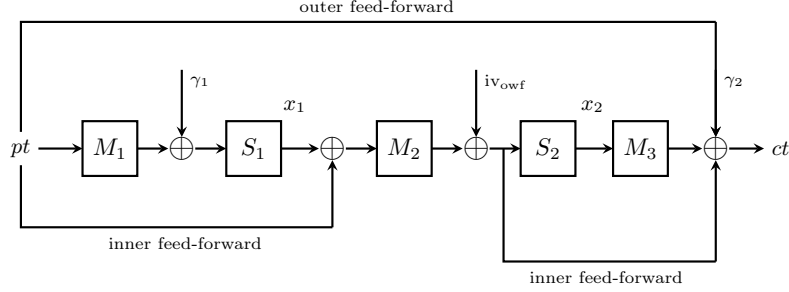


Fig. 1: Overview of the proposed one-way function construction.

where $\gcd(e, n) = 1$. This condition implies $\gcd(2^e - 1, 2^n - 1) = 1$, and therefore the map $x \mapsto x^{2^e - 1}$ is a permutation over $\mathbb{F}_{2^n}^*$. Consequently, the inverse exponent $\bar{e} = (2^e - 1)^{-1} \bmod (2^n - 1)$ defines a permutation over $\mathbb{F}_{2^n}^*$. This choice prevents a high abundance of low-degree relations while avoiding potential overdefined algebraic structures.

Linear Components.

Pylon includes three types of linear components: round constants γ_i , linear diffusion layer $M_i \in (\mathbb{F}_2)^{n \times n}$, and feed-forward structures. To facilitate the implementation and analysis, M_i can be equivalently represented as a linearized polynomial $L_i \in \mathbb{F}_{2^n}[X]$ of the form $L_i(X) = \sum_{j=0}^{n-1} a_j^{(i)} X^{2^j}$, where the coefficients $a_j^{(i)} \in \mathbb{F}_{2^n}$. Throughout our construction, we require that $a_j^{(i)} \neq 0$ for all $j \in \{0, \dots, n-1\}$, to ensure that the polynomial in use is of maximum degree and as dense as possible. This design choice aims to maximize diffusion and to avoid sparse linear structures that could potentially facilitate algebraic or linear cryptanalytic attacks.

Remark on Matrix and Round Constants Generations. We adopt a parameter generation method inspired by the RAIN design [17]. For clarity, this section describes the matrices and round constants generation procedure for a single round and omits the round index, and denote M_i and its corresponding linearized polynomial L_i by M and L .

The procedure can be divided into three steps:

1. *Sampling a candidate linear layer L .* We sample the coefficients of a candidate linearized polynomial

$$L(X) = \sum_{j=0}^{n-1} a_j X^{2^j}$$

from a pseudo-random bit stream and parse them as elements of $\mathbb{F}_{2^n}$. If any coefficient a_j is zero, the sample is rejected and resampled, as the design

mandates a dense linear layer. Once accepted, the polynomial L uniquely defines the corresponding linear transformation $M \in (\mathbb{F}_2)^{n \times n}$.

2. *Checking that L is a permutation and that L^{-1} is also dense.* We build the Dickson matrix D_L of L by

$$D_L = \begin{pmatrix} a_0 & a_1 & \cdots & a_{n-1} \\ a_{n-1}^2 & a_0^2 & \cdots & a_{n-2}^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{2^{n-1}} & a_2^{2^{n-1}} & \cdots & a_0^{2^{n-1}} \end{pmatrix}.$$

Then L is a permutation if and only if D_L is invertible [28]. If $\det(D_L) = 0$, we reject and resample. If $\det(D_L) \neq 0$, we compute L^{-1} from cofactors of D_L . This gives a linearized form

$$L^{-1}(X) = \frac{1}{\det(D_L)} \sum_{i=0}^{n-1} \tilde{a}_i X^{2^i},$$

where $\tilde{a}_i$ comes from the cofactor values of D_L . We reject the sample if any $\tilde{a}_i = 0$. So both L and L^{-1} have all nonzero coefficients with degree of 2^{n-1} .

3. *Building the binary matrix M .* After we fix L , we convert it to the matrix $M \in (\mathbb{F}_2)^{n \times n}$ by using a basis and a dual basis as shown in [14,38]. Let $\{\beta_1, \dots, \beta_n\}$ be an $\mathbb{F}_2$ -basis of $\mathbb{F}_{2^n}$ and let $\{\beta'_1, \dots, \beta'_n\}$ be the dual basis such that $\text{Tr}(\beta_i \beta'_j) = \delta_{i,j}$, where $\text{Tr} : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_2$ is the trace map. Then we define the matrix entries by

$$M_{i,j} = \text{Tr}(\beta'_i \cdot L(\beta_j)).$$

This derives the $\mathbb{F}_2$ -linear map used in the round.

Round constants are generated in a similar way from the same pseudo-random stream and parsed as elements of $\mathbb{F}_{2^n}$. We generate the round constants with the SHAKE256 extendable-output function [31]. We initialize SHAKE256 with the UTF-8 string **GreatWall-N** where N is replaced by the state size in bits. For each round i , we squeeze n bits from SHAKE256 and interpret them as an element $\gamma_i \in \mathbb{F}_{2^n}$. We also use the same SHAKE256 stream to sample the values needed for the linear layer generation above, such as the coefficients $a_j \in \mathbb{F}_{2^n}$. These constants are public and fixed, and they are only used to avoid symmetries across rounds. So using SHAKE256 [31] is reasonable, since this step does not need to provide security strength.

Recommended Parameters. The concrete instances and recommended parameters of Pylon are listed in Table 2.

2.3 Specifications of the GreatWall Signature Scheme

VOLE-in-the-Head. VOLE-in-the-Head (VOLEith) [9] is a general framework for transforming VOLE-based zero-knowledge proofs [8,37,41] into publicly verifiable proofs, and hence into digital signatures via the Fiat-Shamir transform

Variant	Parameters (λ, n, e_1, e_2)	Extension Field	Irreducible Polynomial
Pylon-128	(128, 137, 70, 75)	$\mathbb{F}_{2^{137}}$	$X^{137} + X^{21} + 1$
Pylon-192	(192, 197, 94, 105)	$\mathbb{F}_{2^{197}}$	$X^{197} + X^{21} + X^2 + X + 1$
Pylon-256	(256, 263, 129, 136)	$\mathbb{F}_{2^{263}}$	$X^{263} + X^{93} + 1$
Pylon-512	(512, 521, 248, 273)	$\mathbb{F}_{2^{521}}$	$X^{521} + X^{32} + 1$

Table 2: Recommended sets of parameters of Pylon.

[20]. At a high level, the prover first commits to randomly generated VOLE-related data without knowing the final global key Δ , then a challenge determines Δ , and finally the prover opens the commitments in a way that allows the verifier to reconstruct the corresponding VOLE keys and check the proof.

VOLE correlations. Let $\mathbb{F}_{2^k}$ be a finite field. A length- ℓ VOLE correlation consists of values

$$(u_i, v_i, q_i, \Delta) \in \mathbb{F}_2 \times \mathbb{F}_{2^k} \times \mathbb{F}_{2^k} \times \mathbb{F}_{2^k}, \quad i = 0, \dots, \ell - 1,$$

such that $q_i = u_i \cdot \Delta - v_i$. Here, the prover holds (u_i, v_i) , whereas the verifier holds (Δ, q_i) . Since the relation is linear in Δ , VOLE can be viewed as a linearly homomorphic commitment to the bits u_i . This property is the main reason VOLE is well suited for efficient zero-knowledge proofs.

Small-field instantiations and concatenation. In practice, directly working over $\mathbb{F}_{2^\lambda}$ is too expensive. VOLEitH therefore instantiates several smaller VOLE correlations in parallel and concatenates them to obtain a large-field correlation over $\mathbb{F}_{2^\lambda}$ [10]. Concretely, one chooses a repetition parameter τ and defines

$$k_0 = \left\lceil \frac{\lambda}{\tau} \right\rceil, \quad k_1 = \left\lfloor \frac{\lambda}{\tau} \right\rfloor, \quad \tau_0 = \lambda \bmod \tau, \quad \tau_1 = \tau - \tau_0,$$

so that $\tau_0 k_0 + \tau_1 k_1 = \lambda$. Then τ_0 instances over $\mathbb{F}_{2^{k_0}}$ and τ_1 instances over $\mathbb{F}_{2^{k_1}}$ are generated and concatenated column-wise to form one VOLE relation over $\mathbb{F}_{2^\lambda}$.

All-but-one vector commitments from GGM trees. The main tool enabling delayed VOLE generation is an all-but-one vector commitment based on a GGM tree [22, 10]. Starting from a root seed r , a length-doubling pseudorandom generator recursively expands the seed into a complete binary tree with $N = 2^d$ leaves. Each leaf key is hashed to obtain $(\text{sd}_j, \text{com}_j)$, $j = 0, \dots, N - 1$, where sd_j is the leaf seed used later for VOLE generation and com_j is its local commitment. The overall commitment is

$$h_{\text{com}} = H_1(\text{com}_0, \dots, \text{com}_{N-1}).$$

To open all positions except one hidden index j^* , the prover sends the copath of j^* together with com_{j^*} . This allows the verifier to reconstruct all other leaves and recompute the global commitment, while the unopened leaf remains hidden.

From committed seeds to VOLE. Let the leaf seeds be expanded into vectors $r_0, r_1, \dots, r_{N-1} \in \mathbb{F}_2^{\hat{\ell}}$. A small-field VOLE instance is then derived by defining

$$u = \sum_{j=0}^{N-1} r_j, \quad v = \sum_{j=0}^{N-1} j \cdot r_j,$$

where j is interpreted as an element of $\mathbb{F}_{2^k}$. If the verifier later learns all leaf data except at index Δ , it can compute

$$q = \sum_{j=0}^{N-1} (\Delta - j) \cdot r_j = \Delta \cdot \sum_{j=0}^{N-1} r_j - \sum_{j=0}^{N-1} j \cdot r_j = \Delta \cdot u - v.$$

Hence, opening all but one leaf directly yields a VOLE correlation whose challenge value is the unopened index.

Witness masking and chosen VOLE. The VOLE values generated above are initially random and independent of the witness. To bind them to the witness w of the underlying one-way-function relation, the prover masks the witness with the random VOLE secret and publishes $d = w - u$. Equivalently, it writes $w = u + d$. Since the verifier already holds a VOLE key of the form $q_i = u_i \Delta - v_i$, it can update this key using the public mask d_i as

$$q'_i = q_i + d_i \Delta = (u_i \Delta - v_i) + d_i \Delta = (u_i + d_i) \Delta - v_i = w_i \Delta - v_i.$$

Therefore, after applying the mask, the verifier obtains a VOLE authentication of the actual witness bit w_i . In this way, the original random VOLE is transformed into a *chosen VOLE* bound to the witness. This step is crucial, because the subsequent proof system no longer works with a random hidden vector u , but with the witness w itself.

QuickSilver-based proof stage. Once the chosen VOLE correlations for the witness are available, VOLEitH is typically combined with the QuickSilver proof system to prove that the witness satisfies the target circuit [41]. Consider an arithmetic or Boolean circuit C with multiplication gates. For each input wire and each multiplication-gate output wire, the prover and verifier hold corresponding VOLE-authenticated values. Linear operations can be handled locally thanks to the linearity of VOLE, so only multiplication constraints require an explicit check.

For a multiplication gate indexed by (α, β, γ) , where the intended relation is $w_\gamma = w_\alpha w_\beta$, the verifier computes $b_i = q_\alpha q_\beta - q_\gamma \Delta$. Substituting the VOLE relations

$$q_\alpha = w_\alpha \Delta - v_\alpha, \quad q_\beta = w_\beta \Delta - v_\beta, \quad q_\gamma = w_\gamma \Delta - v_\gamma,$$

one obtains

$$\begin{aligned} b_i &= (w_\alpha \Delta - v_\alpha)(w_\beta \Delta - v_\beta) - (w_\gamma \Delta - v_\gamma) \Delta \\ &= v_\alpha v_\beta + (w_\alpha v_\beta + w_\beta v_\alpha - v_\gamma) \Delta + (w_\alpha w_\beta - w_\gamma) \Delta^2. \end{aligned}$$

If the multiplication constraint is satisfied, i.e. $w_\gamma = w_\alpha w_\beta$, then the quadratic term vanishes and b_i becomes an affine function of Δ :

$$b_i = a_{0,i} + a_{1,i}\Delta,$$

where $a_{0,i} = v_\alpha v_\beta$, $a_{1,i} = w_\alpha v_\beta + w_\beta v_\alpha - v_\gamma$. Thus, to prove correctness of the multiplication gate, it suffices for the prover to provide information allowing the verifier to check this affine relation.

To reduce communication, QuickSilver does not send $(a_{0,i}, a_{1,i})$ for every multiplication gate separately. Instead, the verifier samples a random challenge χ , and the prover forms a random linear combination

$$a_0 = \sum_i \chi^i a_{0,i}, \quad a_1 = \sum_i \chi^i a_{1,i},$$

while the verifier computes $b = \sum_i \chi^i b_i$. The verifier then only needs to check the single equality $b = a_0 + a_1\Delta$.

Batch All-but-one Vector Commitment. Let λ be the security parameter and let $\mathcal{M}$ be a message space. For an integer τ , let $(N_0, \dots, N_{\tau-1})$, be the lengths of τ committed vectors. We use $[n] = \{0, \dots, n-1\}$. A batch all-but-one vector commitment (BAVC) scheme [7] is a tuple of algorithms

$$\text{BAVC} = (\text{Commit}, \text{Open}, \text{Verify})$$

defined as follows.

Commit() $\rightarrow (\text{com}, \text{decom}, (m_1^{(\alpha)}, \dots, m_{N_\alpha}^{(\alpha)})_{\alpha \in [\tau]}):$
output a commitment **com** with opening information **decom** for messages

$$(m_1^{(\alpha)}, \dots, m_{N_\alpha}^{(\alpha)})_{\alpha \in [\tau]} \in \mathcal{M}^{N_1 + \dots + N_\tau}.$$

Open(**decom**, I) $\rightarrow \text{decom}_I \vee \perp$:

On input an opening information **decom** and the index vector $I \in [N_1] \times \dots \times [N_\tau]$, output $\perp$ or an opening **decom** _{I} for I . Here, $I = (i_1, \dots, i_\tau)$ specifies the unopened position in each committed vector.

Verify(**com**, **decom** _{I} , I) $\rightarrow ((m_j^{(\alpha)})_{j \in [N_\alpha] \setminus \{i_\alpha\}})_{\alpha \in [\tau]} \vee \perp$:

Given a commitment **com**, an opening **decom** _{I} for an index vector I , either output all messages

$$((m_j^{(\alpha)})_{j \in [N_\alpha] \setminus \{i_\alpha\}})_{\alpha \in [\tau]}$$

and accept the opening, or output $\perp$ and reject the opening.

Correctness with aborts. BAVC is correct with aborts if, for every index vector

$$I \in [N_0] \times \dots \times [N_{\tau-1}],$$

the following experiment outputs true except with negligible probability:

$$(\text{com}, \text{decom}, M) \leftarrow \text{Commit}(),$$

where

$$M = \left(m_0^{(\alpha)}, \dots, m_{N_\alpha-1}^{(\alpha)} \right)_{\alpha \in [\tau]}.$$

For every

$$\text{decom}_I \leftarrow \text{Open}(\text{decom}, I),$$

it holds that either $\text{decom}_I = \perp$, or

$$\text{Verify}(\text{com}, \text{decom}_I, I) = \left(\left(m_j^{(\alpha)} \right)_{j \in [N_\alpha] \setminus \{i_\alpha\}} \right)_{\alpha \in [\tau]}.$$

Thus, whenever the opening does not abort, the verifier can recover all opened messages consistently with the original commitment, while the entries $\left(m_{i_\alpha}^{(\alpha)} \right)_{\alpha \in [\tau]}$ remain unopened.

GGM-based instantiation. In VOLE-in-the-Head based signatures, BAVC is typically instantiated from a GGM tree. The committed messages are pseudorandom seeds derived from a root seed by repeatedly applying a length-doubling pseudorandom generator. Instead of using τ independent trees, the optimized construction uses a single interleaved GGM tree with $L = \sum_{\alpha=0}^{\tau-1} N_\alpha$ leaves. Each pair (α, j) , corresponding to the j -th entry of the α -th vector, is mapped to a unique leaf position

$$\pi(\alpha, j) \in [L - 1, 2L - 2].$$

The leaf at position $\pi(\alpha, j)$ determines the seed $m_j^{(\alpha)}$ and its leaf commitment. The vector-level commitments are obtained by hashing the leaf commitments inside each vector, and the final commitment **com** is obtained by hashing all vector-level commitments.

Given an index vector $I = (i_0, \dots, i_{\tau-1})$, the opening algorithm keeps the τ leaves

$$\pi(0, i_0), \dots, \pi(\tau - 1, i_{\tau-1})$$

unopened and reveals enough internal tree nodes to reconstruct all remaining leaves. Since different opening paths may merge in the single interleaved tree, fewer internal nodes need to be sent compared with opening τ independent GGM trees. To keep the decommitment size fixed, the scheme sets an opening threshold T_{open} . If more than T_{open} internal nodes are required, then $\text{Open}(\text{decom}, I) = \perp$.

Grinding. In the non-interactive signature scheme, the index vector I is derived from a Fiat-Shamir challenge. Let w_{grind} be the grinding parameter. The signer computes

$$\text{chall}_3 = H(\text{transcript} \parallel \text{ctr}) \in \{0, 1\}^\lambda,$$

where ctr is a counter initialized to zero. The first $\lambda - w_{\text{grind}}$ bits of chall_3 are decoded into the index vector I , while the remaining w_{grind} bits are required to be zero. If this condition is not satisfied, or if $\text{Open}(\text{decom}, I) = \perp$, the signer increments ctr and recomputes the challenge. The final counter value is included in the signature, so the verifier can recompute chall_3 , check the grinding condition, decode the same index vector I , and verify the BAVC opening.

Next the whole description of the GreatWall signature scheme will be given, consisting of key generation, signing, and verification algorithms.

2.3.1 Key Generation.

The key generation takes as input a randomly generated pt and outputs a public key $(\text{iv}_{\text{owf}}, ct)$ and a secret key pt such that $ct = \text{Pylon}(\text{iv}_{\text{owf}}, pt)$. The algorithm is given in algorithm 1.

Algorithm 1: $\text{Pylon.KeyGen}(1^\lambda)$

```

1 repeat
2    $pt \xleftarrow{\$} \text{GF}(2^n)$ 
3    $\text{iv}_{\text{owf}} \xleftarrow{\$} \{0, 1\}^n$ 
4    $\alpha_1 := \text{SHAKE}(\text{iv}_{\text{owf}} \| 0)$ 
5    $\alpha_2 := \text{SHAKE}(\text{iv}_{\text{owf}} \| 1)$ 
6    $(ct, x_1, x_2) := \text{Pylon}(\text{iv}_{\text{owf}}, pt)$ 
7 until  $\langle \alpha_1, x_1 \rangle \neq 0 \wedge \langle \alpha_2, x_2 \rangle \neq 0$ 
8  $pk := (\text{iv}_{\text{owf}}, ct)$ 
9  $sk := pt$ 
10 return  $(pk, sk)$ 
```

2.3.2 Signature Signing.

The signing algorithm of the GreatWall signature is given in algorithm 2. It takes as input the secret and public keys $(pt, (\text{iv}_{\text{owf}}, ct))$ and a message m and outputs a signature σ . The signing algorithm utilizes several procedures defined in earlier sections, and the signature generation is executed through four main phases, which are delineated by the derivation of these challenges.

Algorithm 2: GreatWall.Sign(m,pk,sk)

```

1  $pt \leftarrow sk$ 
2  $iv_{owf}, ct \leftarrow pk$ 
3  $ctr \leftarrow 0$  // 32-bit int
4  $\rho \xleftarrow{\$} \{0, 1\}^\lambda$  // Signature randomness
5  $\mu := H_1(ct \| m; 2\lambda)$ 
6  $(r, iv_{vole}) := H_3(pt \| \mu \| \rho)$  //  $r \in \{0, 1\}^\lambda, iv_{vole} \in \{0, 1\}^{128}$ 
7 // Commitment and VOLE Derandomization
8  $(com, decom, c_1, \dots, c_{\tau-1}, \mathbf{u}, \mathbf{V}) := \text{VOLECommit}(r, iv_{vole}, \ell + 2\lambda + B)$ 
9  $chall_1 := H_2^1(\mu \| com \| c_1 \| \dots \| c_{\tau-1} \| iv_{vole}; 5\lambda + 64)$ 
10 // VOLE consistency check and witness commitment
11  $\tilde{\mathbf{u}} := \text{VOLEHash}(chall_1, \mathbf{u}) \in \{0, 1\}^{\lambda+B}$ 
12  $\tilde{\mathbf{V}} := \text{VOLEHash}(chall_1, \mathbf{V}) \in \{0, 1\}^{(\lambda+B) \times \lambda}$ 
13  $h_V := H_1(\tilde{\mathbf{V}})$  // hash column-wise
14  $(ct, x_1, x_2) := \text{Pylon}(iv_{owf}, pt)$ 
15  $w := (pt, x_1)$ 
16  $\mathbf{d} := \mathbf{w} \oplus \mathbf{u}_{[0..\ell]}$  // Mask extended witness
17  $chall_2 := H_2^2(chall_1 \| \tilde{\mathbf{u}} \| h_V \| \mathbf{d}; 3\lambda + 64)$ 
18 // Pylon proof
19  $\mathbf{u} := \mathbf{u}_{[0..\ell+\lambda]}$ 
20  $\mathbf{V} := \mathbf{V}_{[0..\ell+\lambda]}$  // drop last  $\lambda + B$  rows
21  $(\tilde{a}, \tilde{b}) := \text{ZK.OWFPProve}(\mathbf{w}, \mathbf{u}, \mathbf{V}, ct, iv_{owf}, chall_2)$ 
22  $ctr \leftarrow -1$ 
23 repeat
24    $ctr \leftarrow ctr + 1$ 
25    $chall_3 \leftarrow H_2^3(chall_2 \| \tilde{a} \| \tilde{b} \| ctr; \lambda)$ 
26   if  $chall_3[\lambda - w_{grind} : \lambda] \neq 0^{w_{grind}}$  then
27     continue
28    $I \leftarrow \text{DecodeAllChall3}(chall_3[0 : \lambda - w_{grind}])$ 
29    $decom_I \leftarrow \text{BAVC.Open}(decom, I)$ 
30 until  $decom_I \neq \perp$ 
31 // Build VOLE challenge and open Vector Commitments
32 return  $\sigma := ((c_i)_{i \in [1..\tau]}, \tilde{\mathbf{u}}, \mathbf{d}, \tilde{a}, decom_I, chall_3, iv_{vole}, ctr)$ 

```

Phase 1. The signing procedure begins by setting an integer counter ctr to zero, which will be utilized later during the “grind check”. Following this, the algorithm samples a random value $\rho \in \{0, 1\}^\lambda$ to serve as the signature randomness. Next, it hashes the public key pk and the message m together via H_1 to output a 2λ -bit digest μ . The signer then processes the secret key pt , the digest μ , and ρ through H_3 . This operation generates a λ -bit PRG seed r alongside an intermediate 128-bit string iv_{vole} . Using the pair (r, iv_{vole}) , the signer calls the `VOLECommit` function to simultaneously commit to all VOLE instances. This single execution packages the small-VOLE vectors into a comprehensive GGM-

based batch vector commitment. It outputs a primary commitment com , a bulk decommitment structure decom that allows opening most seeds while keeping exactly one hidden per vector, a series of correction vectors $c_1, \dots, c_{\tau-1}$ designed to align every VOLE instance to a shared secret, and the global VOLE secret u along with the MAC tags V . Finally, the signer extracts the first challenge chall_1 by hashing μ , com , the correction vectors $\{c_i\}$, and iv_{vole} via H_2^1 , requesting an output of $5\lambda + 64$ bits.

Algorithm 3: $\text{VOLECommit}(r, \text{iv}_{\text{vole}}, \widehat{\ell})$

```

1 // Set VOLE vector sizes
2 for  $i \in [0..\tau)$  do
3    $\kappa_i := \begin{cases} k_0, & i < \tau_0, \\ k_1, & i \geq \tau_0 \end{cases}$ 
4    $N_i := 2^{\kappa_i}$ 
5 // Batch all-but-one vector commitment
6  $(\text{com}, \text{decom}, (\text{sd}_{i,0}, \dots, \text{sd}_{i,N_i-1})_{i \in [0..\tau)}) :=$ 
    $\text{BAVC.Commit}(r, \text{iv}_{\text{vole}}, (N_i)_{i \in [0..\tau)})$ 
7 // Convert committed seeds to small VOLE correlations
8 for  $i \in [0..\tau)$  do
9    $(\mathbf{u}_i, \mathbf{v}_{i,0}, \dots, \mathbf{v}_{i,\kappa_i-1}) := \text{ConvertToVOLE}(\text{sd}_{i,0}, \dots, \text{sd}_{i,N_i-1}, \text{iv}_{\text{vole}}, \widehat{\ell})$ 
10   $\mathbf{V}_i := [\mathbf{v}_{i,0} \ \dots \ \mathbf{v}_{i,\kappa_i-1}] \in \mathbb{F}_2^{\widehat{\ell} \times \kappa_i}$ 
11 // Concatenate small VOLE tags into one large VOLE tag
   matrix
12  $\mathbf{V} := [\mathbf{V}_0 \ \dots \ \mathbf{V}_{\tau-1}] \in \mathbb{F}_2^{\widehat{\ell} \times \lambda}$ 
13 // Use the first VOLE secret as the common secret
14  $\mathbf{u} := \mathbf{u}_0$ 
15 // Correction values for enforcing the same VOLE secret
16 for  $i \in [1..\tau)$  do
17    $c_i := \mathbf{u} \oplus \mathbf{u}_i$ 
18 return  $(\text{com}, \text{decom}, c_1, \dots, c_{\tau-1}, \mathbf{u}, \mathbf{V})$ 

```

Phase 2. In the second phase, the signer computes $\tilde{V}$ and $\tilde{u}$ to prepare for the VOLE consistency check. This is achieved by invoking VOLEHash on inputs (V, chall_1) and (u, chall_1) , respectively. Then h_V is computed to generate the second challenge. Subsequently, the algorithm runs ExtendWitness using pt and the output of the first S-box to construct the extended witness w . To properly commit to w , the signer computes the difference $d := w \oplus u_{[0..l)}$, effectively masking w with a one-time pad derived from u . A second challenge chall_2 , is then derived by hashing the concatenation of chall_1 , $\tilde{u}$, h_V , and d .

Phase 3. During the third phase, the signer generates the QuickSilver proof. Initially, it truncates u and V to the first $l + \lambda$ rows. It then executes the zero-knowledge proof procedure ZK.OWFProve with inputs w , u , V , ct , and chall_2 .

This function returns $\tilde{a}$, and $\tilde{b}$. The signer then enters a grinding loop: it computes the third challenge $\text{chall}_3 \leftarrow H_2^3(\text{chall}_2 \parallel \tilde{a} \parallel \tilde{b} \parallel \text{ctr}; \lambda)$, and checks if the last w_{grind} bits of chall_3 are zero. If not, it increments ctr and repeats. Once a valid chall_3 is found, the signer decodes the first $\lambda - w_{\text{grind}}$ bits of chall_3 into an index vector I using `DecodeAllChall3`, and calls `BAVC.Open` with the decommitment decom and index I to generate a partial decommitment decom_I . If `BAVC.Open` returns $\perp$ (the opening size exceeds the threshold T_{open}), the loop increments ctr and restarts the challenge computation.

Algorithm 4: `BAVC.Commit( $r, \text{iv}_{\text{vole}}, (N_i)_{i \in [0..\tau]}$ )`

```

1 // Compute the number of leaves in the single batched tree
2  $L := \sum_{i=0}^{\tau-1} N_i$ 
3  $N_{\max} := \max_{i \in [0..\tau)} N_i$ 
4 // Define the interleaving between virtual commitments and
  tree leaves
5  $t := 0$ 
6 for  $j \in [0..N_{\max})$  do
7   for  $i \in [0..\tau)$  do
8     if  $j < N_i$  then
9        $\pi(t) := (i, j)$ 
10       $\pi^{-1}(i, j) := t$ 
11       $t := t + 1$ 
12 // Expand one GGM tree with  $L$  leaves
13  $K_0 := r$ 
14 for  $a \in [0..L - 1)$  do
15    $(K_{2a+1}, K_{2a+2}) := \text{PRG}(K_a \parallel \text{iv}_{\text{vole}}; 2\lambda)$ 
16  $\mathcal{T} := (K_0, K_1, \dots, K_{2L-2})$ 
17 // Derive committed seeds and leaf commitments
18 for  $t \in [0..L)$  do
19    $(i, j) := \pi(t)$ 
20    $(\text{sd}_{i,j}, \text{leafcom}_{i,j}) := H_0(K_{L-1+t} \parallel \text{iv}_{\text{vole}})$ 
21 for  $i \in [0..\tau)$  do
22    $h_i := H_1(\text{leafcom}_{i,0} \parallel \text{leafcom}_{i,1} \parallel \dots \parallel \text{leafcom}_{i,N_i-1})$ 
23  $\text{hcom} := H_1(h_0 \parallel h_1 \parallel \dots \parallel h_{\tau-1})$ 
24  $\text{decom} := (\mathcal{T}, (\text{leafcom}_{i,j})_{i \in [0..\tau), j \in [0..N_i)})$ 
25 return  $(\text{hcom}, \text{decom}, (\text{sd}_{i,0}, \dots, \text{sd}_{i,N_i-1})_{i \in [0..\tau)})$ 

```

Phase 4. In the final phase, the signer assembles the complete signature once the grinding check and partial commitment opening both succeed. The signature σ is constructed as a tuple containing: the correction vectors $c_1, \dots, c_{\tau-1}$, the hashed VOLE secret $\tilde{\mathbf{u}}$, the masked witness $\mathbf{d}$, the zero-knowledge proof tags $\tilde{a}$, the partial decommitment decom_I , the final challenge chall_3 , the intermediate

Algorithm 5: ConvertToVOLE($\text{sd}_0, \dots, \text{sd}_{N-1}, \text{iv}_{\text{vole}}, \widehat{\ell}$)

```

1  $d := \log_2 N$  //  $N = 2^d$ 
2 // Expand committed seeds into length- $\widehat{\ell}$  bit vectors
3 for  $i \in [0..N)$  do
4    $\mathbf{r}_{0,i} := \text{PRG}(\text{sd}_i, \text{iv}_{\text{vole}}; \widehat{\ell})$ 
5 // Initialize the  $d$  VOLE tag columns
6 for  $j \in [0..d)$  do
7    $\mathbf{v}_j := 0^{\widehat{\ell}}$ 
8 // Compress  $N$  expanded vectors into one secret vector and  $d$  tag
   columns
9 for  $j \in [0..d)$  do
10   for  $i \in [0..N/2^{j+1})$  do
11      $\mathbf{v}_j := \mathbf{v}_j \oplus \mathbf{r}_{j,2i+1}$ 
12      $\mathbf{r}_{j+1,i} := \mathbf{r}_{j,2i} \oplus \mathbf{r}_{j,2i+1}$ 
13  $\mathbf{u} := \mathbf{r}_{d,0}$ 
14 return  $(\mathbf{u}, \mathbf{v}_0, \dots, \mathbf{v}_{d-1})$ 

```

initialization vector iv_{vole} , and the grinding counter ctr . The algorithm outputs this tuple σ as the valid signature on message m .

Algorithm 6: VOLEHash(chall, x)

```

1 //  $\mathbf{x} \in \{0, 1\}^{\ell+2\lambda+B}$ 
2  $\mathbf{x}_0 := \mathbf{x}_{[0..\ell+\lambda)}$ 
3  $\mathbf{x}_1 := \mathbf{x}_{[\ell+\lambda..\ell+2\lambda+B)}$ 
4  $(r_0, r_1, r_2, r_3, s, t) := \text{chall}$ 
5 //  $r_0, r_1, r_2, r_3, s \in \{0, 1\}^\lambda, t \in \{0, 1\}^{64}$ 
6 for  $i \in [0..4)$  do
7    $r_i := \text{ToField}(r_i, \lambda)$ 
8  $s := \text{ToField}(s, \lambda)$ 
9  $t := \text{ToField}(t, 64)$ 
10  $\ell' := \lambda \cdot \left\lceil \frac{\ell + \lambda}{\lambda} \right\rceil$ 
11  $\mathbf{x}_0 := \mathbf{x}_0 \| 0^{\ell' - (\ell + \lambda)}$ 
12  $\hat{\mathbf{y}} := \text{ToField}(\mathbf{x}_0, \lambda)$ 
13  $\bar{\mathbf{y}} := \text{ToField}(\mathbf{x}_0, 64)$ 
14  $h_0 := 0$ 
15 for  $i \in [0..\ell'/\lambda)$  do
16    $h_0 := h_0 + s^{\ell'/\lambda - 1 - i} \cdot \hat{\mathbf{y}}[i]$ 
17  $h_1 := 0$ 
18 for  $i \in [0..\ell'/64)$  do
19    $h_1 := h_1 + t^{\ell'/64 - 1 - i} \cdot \bar{\mathbf{y}}[i]$ 
20  $h'_1 := \text{ToField}(\text{ToBits}(h_1) \| 0^{\lambda - 64}, \lambda)$ 
21  $h_2 := r_0 h_0 + r_1 h'_1$ 
22  $h_3 := r_2 h_0 + r_3 h'_1$ 
23  $h := (\text{ToBits}(h_2) \| \text{ToBits}(h_3)_{[0..B)}) \oplus \mathbf{x}_1$ 
24 return  $h$ 

```

Algorithm 7: ZK.OWFProve($(\mathbf{w}, \mathbf{u}, \mathbf{V}, ct, \text{iv}_{\text{owf}}, \text{chall}_2)$)

```

1 for  $i \in [0..\ell]$  do
2    $w[i] \leftarrow \text{ToField}(w[i]; 1)$ 
3 for  $i \in [0..\ell + \lambda]$  do
4    $v[i] \leftarrow \text{ToField}(V[i]; \lambda)$ 
5 Parse  $w[i]_{i \in [0..n]}, v[i]_{i \in [0..n]}$  as  $pt.value, pt.mac$ 
6 Parse  $w[i]_{i \in [n..2n]}, v[i]_{i \in [n..2n]}$  as  $x_1.value, x_1.mac$ 
7  $(\alpha_0^{dot}, \alpha_1^{dot}) \leftarrow \text{SHAKE}(\text{iv}_{\text{owf}}||0), \text{SHAKE}(\text{iv}_{\text{owf}}||1)$ 
8  $((\alpha_1, \beta_1, \gamma_1), (\alpha_2, \beta_2, \gamma_2), x_2) \leftarrow \text{LinearTransform}(pt, x_1, \text{iv}_{\text{owf}}, ct)$ 
9 for  $j \in [1..2]$  do
10   $(A_{0,j}, A_{1,j}) \leftarrow \text{MulCstrnts}(\alpha_j, \beta_j, \gamma_j)$ 
11  $(A_{0,3}, A_{1,3}) \leftarrow \text{DotCstrnt}(\alpha_0^{dot}, x_1)$ 
12  $(A_{0,4}, A_{1,4}) \leftarrow \text{DotCstrnt}(\alpha_1^{dot}, x_2)$ 
13  $a_0 \leftarrow A_{0,1} || A_{0,2} || A_{0,3} || A_{0,4}$ 
14  $a_1 \leftarrow A_{1,1} || A_{1,2} || A_{1,3} || A_{1,4}$ 
15  $u^* \leftarrow \sum_{i \in [0..\lambda]} u[\ell + i] \alpha_\lambda^i, \quad v^* \leftarrow \sum_{i \in [0..\lambda]} v[\ell + i] \alpha_\lambda^i \quad // \text{ where}$ 
     $\alpha_\lambda \in \mathbb{F}_{2^\lambda}$ 
16  $\tilde{a} \leftarrow \text{ZKHash}(\text{chall}_2, a_1 || u^*)$ 
17  $\tilde{b} \leftarrow \text{ZKHash}(\text{chall}_2, a_0 || v^*)$ 
18 return  $(\tilde{a}, \tilde{b})$ 

```

Algorithm 8: LinearTransform($pt, x_1, \text{iv}_{\text{owf}}, ct$)

```

1 // First S-box
2  $s_1 := M_1 \cdot pt \oplus \gamma_1$ 
3  $y_1 := x_1$ 
4  $z_1 := y_1^{2^{e_1}}$ 
5 // Second S-box input
6  $t := y_1 \oplus pt$ 
7  $s_2 := M_2 \cdot t \oplus \text{iv}_{\text{owf}}$ 
8  $y_2 := M_3^{-1} \cdot (ct \oplus s_2 \oplus \gamma_2 \oplus pt)$ 
9  $z_2 := y_2^{2^{e_2}}$ 
10 return  $((s_1, y_1, z_1), (s_2, y_2, z_2), y_2)$ 

```

Algorithm 9: MulCstrnts(x, y, z)

```

1 // x,y,z are represented bitwise
2 // Each bit wire  $a$  has  $a.value \in \mathbb{F}_2$  and  $a.mac \in \mathbb{F}_{2^\lambda}$ 
3 for  $i \in [0..n)$  do
4    $x_i^{(0)} := x_i.mac$ 
5    $x_i^{(1)} := x_i.mac \oplus x_i.value \cdot 1_\lambda$ 
6    $y_i^{(0)} := y_i.mac$ 
7    $y_i^{(1)} := y_i.mac \oplus y_i.value \cdot 1_\lambda$ 
8 for  $i \in [0..n)$  do
9   for  $j \in [0..n)$  do
10     $P_{i+j}^{(0)} := P_{i+j}^{(0)} \oplus x_i.mac \cdot y_j.mac$ 
11     $P_{i+j}^{(1)} := P_{i+j}^{(1)} \oplus (x_i.mac \oplus x_i.value) \cdot (y_j.mac \oplus y_j.value)$ 
12  $P = P \bmod f(X)$ 
13 for  $k \in [0..n)$  do
14    $A_{0,k} := P_k^{(0)} \oplus z_k.value$ 
15    $A_{1,k} := A_{0,k} \oplus P_k^{(1)} \oplus z_k.mac$ 
16  $A_0 := A_{0,0} \| A_{0,1} \| \dots \| A_{0,n-1}$ 
17  $A_1 := A_{1,0} \| A_{1,1} \| \dots \| A_{1,n-1}$ 
18 return  $(A_0, A_1)$ 

```

Algorithm 10: DotCstrnt(α, x)

```

1  $d.value \leftarrow 0$ 
2  $d.mac \leftarrow 0$ 
3 for  $i \in [0..n)$  do
4   if  $\alpha_i = 1$  then
5      $d.value \leftarrow d.value \oplus x_i.value$ 
6      $d.mac \leftarrow d.mac \oplus x_i.mac$ 
7  $A_0 \leftarrow 0$ 
8  $A_1 \leftarrow d.mac$ 
9 return  $(A_0, A_1)$ 

```

Algorithm 11: ZKHash(chall, $(\mathbf{x}_0, x_1)$)

```

1 //  $\mathbf{x}_0 = (x_{0,0}, \dots, x_{0,m-1}) \in \mathbb{F}_{2^\lambda}^m$ 
2 //  $x_1 \in \mathbb{F}_{2^\lambda}$ 
3 //  $\text{chall} \in \{0, 1\}^{3\lambda+64}$ 
4  $(r_0, r_1, s, t) := \text{chall}$ 
5                                     //  $r_0, r_1, s \in \{0, 1\}^\lambda, t \in \{0, 1\}^{64}$ 
6  $r_0 := \text{ToField}(r_0, \lambda)$ 
7  $r_1 := \text{ToField}(r_1, \lambda)$ 
8  $s := \text{ToField}(s, \lambda)$ 
9  $t := \text{ToField}(t \| 0^{\lambda-64}, \lambda)$ 
10  $h_0 := 0$ 
11 for  $i \in [0..m)$  do
12    $h_0 := h_0 + s^{m-1-i} \cdot x_{0,i}$ 
13  $h_1 := 0$ 
14 for  $i \in [0..m)$  do
15    $h_1 := h_1 + t^{m-1-i} \cdot x_{0,i}$ 
16  $h := \text{ToBits}(r_0 h_0 + r_1 h_1 + x_1)$ 
17 return  $h$ 

```

Algorithm 12: BAVC.Open(decom, I)

```

1 Parse decom = ( $\mathcal{T}$ , (leafcom $_{i,j}$ ) $_{i \in [0..\tau], j \in [0..N_i]}$ )
2 Parse  $\mathcal{T} = (K_0, K_1, \dots, K_{2L-2})$ 
3  $N_{\max} := \max_{i \in [0..\tau]} N_i$ 
4  $t := 0$ 
5 for  $j \in [0..N_{\max}]$  do
6   for  $i \in [0..\tau]$  do
7     if  $j < N_i$  then
8        $\pi(t) := (i, j)$ 
9        $\pi^{-1}(i, j) := t$ 
10       $t := t + 1$ 
11  $\mathcal{H} := \emptyset$ 
12 for  $i \in [0..\tau]$  do
13    $t_i := \pi^{-1}(i, I_i)$ 
14    $\mathcal{H} := \mathcal{H} \cup \{L - 1 + t_i\}$ 
15  $\mathcal{S} := \emptyset$ 
16 for  $t \in [0..L]$  do
17   if  $L - 1 + t \notin \mathcal{H}$  then
18      $\mathcal{S} := \mathcal{S} \cup \{L - 1 + t\}$ 
19 for  $p = L - 2, L - 3, \dots, 0$  do
20   if  $2p + 1 \in \mathcal{S}$  and  $2p + 2 \in \mathcal{S}$  then
21      $\mathcal{S} := (\mathcal{S} \setminus \{2p + 1, 2p + 2\}) \cup \{p\}$ 
22 if  $|\mathcal{S}| > T_{\text{open}}$  then
23   return  $\perp$ 
24  $\mathcal{O} := \emptyset$ 
25 foreach  $p \in \mathcal{S}$  do
26    $\mathcal{O} := \mathcal{O} \cup \{(p, K_p)\}$ 
27 for  $i \in [0..\tau]$  do
28   hleaf $_i := \text{leafcom}_{i, I_i}$ 
29 decom $_I := (\mathcal{O}, (\text{hleaf}_0, \dots, \text{hleaf}_{\tau-1}))$ 
30 return decom $_I$ 

```

2.3.3 Signature Verification.

The verification algorithm of the GreatWall signature is given in algorithm 13. It takes as input the public key pk , the message m and the signature σ and outputs either Accept or Reject.

Phase 1. The verifier first parses the signature into its individual components. It recomputes $\mu := H_1(ct \| m; 2\lambda)$ from the public key and the message.

Before reconstructing the VOLE transcript, the verifier checks whether the grinding condition is satisfied. Specifically, it verifies that the last w_{grind} bits of

chall_3 are all zero:

$$\text{chall}_3[\lambda - w_{\text{grind}} : \lambda] = 0^{w_{\text{grind}}}.$$

If this condition fails, the verifier immediately rejects the signature.

Phase 2. Assuming the grinding check succeeds, the verifier reconstructs the VOLE receiver side from the partial decommitment. It invokes

$$(\text{com}, Q) \leftarrow \text{VOLEReconstruct}(\text{decom}_I, \text{chall}_3[0..\lambda - w_{\text{grind}}], c_1, \dots, c_{\tau-1}, \text{iv}_{\text{vole}}).$$

This procedure uses the index information encoded in chall_3 to reconstruct all opened GGM leaves, recompute the corresponding batch commitment com , and recover the verifier's VOLE matrix Q . If the reconstruction algorithm returns $\perp$, the verifier rejects. Once com and Q have been obtained, the verifier recomputes the first Fiat–Shamir challenge by hashing the message digest, the reconstructed commitment, the correction vectors, and iv_{vole} :

$$\text{chall}_1 := H_2^1(\mu \|\text{com}\|c_1\| \dots \|c_{\tau-1}\|\text{iv}_{\text{vole}}; 5\lambda + 64).$$

Phase 3. The verifier next performs the VOLE consistency check. It first uses a loop to recover the real Q with the correction vectors c_i . Then it compresses the reconstructed VOLE matrix using the universal hash determined by chall_1 :

$$\tilde{Q} := \text{VOLEHash}(\text{chall}_1, Q).$$

Then it parses the bits of chall_3 as

$$\text{chall}_3 = (\delta_0, \dots, \delta_{\lambda-1}) \in \{0, 1\}^\lambda.$$

Using these bits and the signature component $\tilde{u}$, the verifier removes the challenge-dependent VOLE offset from $\tilde{Q}$ and obtains

$$h_V := H_1(\tilde{Q} \oplus [\delta_0 \cdot \tilde{u} \parallel \delta_1 \cdot \tilde{u} \parallel \dots \parallel \delta_{\lambda-1} \cdot \tilde{u}]).$$

The verifier then recomputes the second Fiat–Shamir challenge as

$$\text{chall}_2 := H_2^2(\text{chall}_1 \|\tilde{u}\|h_V\|d; 3\lambda + 64),$$

where D is encoded in column-major order.

Phase 4. Finally, the verifier checks the QuickSilver proof and the consistency of the final Fiat–Shamir challenge. It runs the verification procedure for the OWF proof:

$$\tilde{b} \leftarrow \text{ZK.OWFVerify}(d, Q[0..\ell + \lambda], \text{chall}_2, \text{chall}_3, \tilde{a}, ct).$$

This step recomputes the missing proof component $\tilde{b}$ from the witness commitment, the reconstructed VOLE information, and the public OWF instance. The verifier then recomputes $\text{chall}'_3 := H_2^3(\text{chall}_2 \|\tilde{a}\|\tilde{b}\|\text{ctr}; \lambda)$. The signature is accepted if and only if $\text{chall}'_3 = \text{chall}_3$. Otherwise, the verifier rejects.

Algorithm 13: GreatWall.Verify(m, pk, σ)

```

1 Parse  $\sigma = ((c_i)_{i \in [1..\tau)}, \tilde{\mathbf{u}}, \mathbf{d}, \tilde{a}, \text{decom}_I, \text{chall}_3, \text{iv}_{\text{vole}}, \text{ctr})$ 
2  $\text{iv}_{\text{owf}}, ct \leftarrow pk$ 
3  $\mu \leftarrow H_1(ct \| m)$ 
4 if  $\text{chall}_3[\lambda - w_{\text{grind}} : \lambda] \neq 0^{w_{\text{grind}}}$  then
5   return false
6 // Reconstruct VOLEs and check commitment
7  $(\text{com}, Q') \leftarrow \text{VOLEReconstruct}(\text{decom}_I, \text{chall}_3[0 : \lambda - w_{\text{grind}}], c_1, \dots, c_{\tau-1}, \text{iv}_{\text{vole}})$ 
8 if  $\text{VOLEReconstruct}$  returned  $\perp$  then
9   return false
10  $\text{chall}_1 \leftarrow H_2^1(\mu \| \text{com} \| c_1 \| \dots \| c_{\tau-1} \| \text{iv}_{\text{vole}}; 5\lambda + 64)$ 
11 // Apply the VOLE correction values and check consistency
12  $Q_0 \leftarrow Q'_0$ 
13 for  $i \in [0..\tau)$  do
14    $b \leftarrow 0$  if  $i < \tau_0$  else 1
15    $(\delta_0, \dots, \delta_{k_b-1}) \leftarrow \text{ChalDec}(\text{chall}_3, i; \text{param})$ 
16    $\tilde{D}_i \leftarrow [\delta_0 \cdot \tilde{u} \| \dots \| \delta_{k_b-1} \cdot \tilde{u}] \in \{0, 1\}^{(\lambda+B) \times k_b}$ 
17   if  $i > 0$  then
18      $Q_i \leftarrow Q'_i \oplus [\delta_0 \cdot c_i \| \dots \| \delta_{k_b-1} \cdot c_i]$ 
19  $Q \leftarrow [Q_0 \dots Q_{\tau-1}]$ 
20  $\tilde{Q} \leftarrow \text{VOLEHash}(\text{chall}_1, Q) \in \{0, 1\}^{(\lambda+B) \times \lambda}$ 
21  $h_V \leftarrow H_1(\tilde{Q} \oplus [\tilde{D}_0 \dots \tilde{D}_{\tau-1}])$ 
22  $\text{chall}_2 \leftarrow H_2^2(\text{chall}_1 \| \tilde{\mathbf{u}} \| h_V \| \mathbf{d}; 3\lambda + 64)$ 
23 // Compute Pylon consistency values
24  $\tilde{b} \leftarrow \text{ZK.OWFVerify}(\mathbf{d}, \mathbf{Q}_{[0..\ell+\lambda)}, \text{chall}_2, \text{chall}_3, \tilde{a}, ct, \text{iv}_{\text{owf}})$ 
25 Compute  $\text{chall}'_3 \leftarrow H_2^3(\text{chall}_2 \| \tilde{a} \| \tilde{b} \| \text{ctr}; \lambda)$ 
26 return true if  $\text{chall}_3 = \text{chall}'_3$  else false

```

Algorithm 14: $\text{VOLEReconstruct}(\text{chall}_3, \text{decom}_I, \text{iv}_{\text{vole}}, \widehat{\ell})$

```

1 if  $\text{chall}_3[\lambda - w_{\text{grind}}.. \lambda] \neq 0^{w_{\text{grind}}}$  then
2   return  $\perp$ 
3  $\text{pos} := 0$ 
4 for  $i \in [0..\tau)$  do
5    $\kappa_i := \begin{cases} k_0, & i < \tau_0, \\ k_1, & i \geq \tau_0 \end{cases}$ 
6    $N_i := 2^{\kappa_i}$ 
7    $I_i := \sum_{s=0}^{\kappa_i-1} \text{chall}_3[\text{pos} + s] \cdot 2^s$ 
8    $\text{pos} := \text{pos} + \kappa_i$ 
9 if  $\text{pos} \neq \lambda - w_{\text{grind}}$  then
10  return  $\perp$ 
11 Parse  $\text{decom}_I = (\mathcal{O}, (\text{hleaf}_0, \dots, \text{hleaf}_{\tau-1}))$ 
12  $L := \sum_{i=0}^{\tau-1} N_i$ 
13  $N_{\max} := \max_{i \in [0..\tau)} N_i$ 
14  $t := 0$ 
15 for  $j \in [0..N_{\max})$  do
16   for  $i \in [0..\tau)$  do
17     if  $j < N_i$  then
18        $\pi(t) := (i, j)$ 
19        $\pi^{-1}(i, j) := t$ 
20        $t := t + 1$ 
21 for  $p \in [0..2L - 1)$  do
22    $K_p := \perp$ 
23 foreach  $(p, K) \in \mathcal{O}$  do
24    $K_p := K$ 
25 for  $p \in [0..L - 1)$  do
26   if  $K_p \neq \perp$  then
27      $(K_{2p+1}, K_{2p+2}) := \text{PRG}(K_p, \text{iv}_{\text{vole}}; 2\lambda)$ 
28 for  $t \in [0..L)$  do
29    $(i, j) := \pi(t)$ 
30   if  $j = I_i$  then
31      $\text{sd}_{i,j} := \perp$ 
32      $\text{leafcom}_{i,j} := \text{hleaf}_i$ 
33   else
34     if  $K_{L-1+t} = \perp$  then
35       return  $\perp$ 
36      $(\text{sd}_{i,j}, \text{leafcom}_{i,j}) := H_0(K_{L-1+t} \parallel \text{iv}_{\text{vole}})$ 
37 for  $i \in [0..\tau)$  do
38    $h_i := H_1(\text{leafcom}_{i,0} \parallel \dots \parallel \text{leafcom}_{i,N_i-1})$ 
39  $\text{hcom} := H_1(h_0 \parallel h_1 \parallel \dots \parallel h_{\tau-1})$ 
40 for  $i \in [0..\tau)$  do
41    $\Delta_i := I_i$ 
42    $\text{sd}'_{i,0} := \perp$ 
43   for  $j \in [1..N_i)$  do
44      $\text{sd}'_{i,j} := \text{sd}_{i,j \oplus \Delta_i}$ 
45    $(u_i^0, q_{i,0}, \dots, q_{i,\kappa_i-1}) := \text{ConvertToVOLE}(\text{sd}'_{i,0}, \dots, \text{sd}'_{i,N_i-1}, \text{iv}_{\text{vole}}, \widehat{\ell})$ 
46    $Q_i^0 := [q_{i,0} \ \dots \ q_{i,\kappa_i-1}] \in \mathbb{F}_2^{\widehat{\ell} \times \kappa_i}$ 
47 return  $(\text{hcom}, Q_0^0, \dots, Q_{\tau-1}^0)$ 

```

Algorithm 15: ZK.OWFVerify(($\mathbf{d}, \mathbf{Q}, \text{chall}_2, \text{chall}_3, \tilde{a}, ct, \text{iv}_{\text{owf}}$))

```

1  $\Delta := \text{ToField}(\text{chall}_3)$ 
2 Write  $Q := [Q_0 \cdots Q_{\tau-1}]$  //  $Q_0, \dots, Q_{\tau-1}$  have  $k_0$  columns,
   others  $k_1$ 
3 for  $i \in [0.. \tau)$  do
4    $b \leftarrow 0$  if  $i < \tau_0$  else 1
5    $(\delta_0, \dots, \delta_{k_b-1}) \leftarrow \text{ChalDec}(\text{chall}_3, i; \text{param})$ 
6    $Q_i \leftarrow [q_{i,0}, \dots, q_{i,k_b-1}]$  // consider  $k_b$  columns separately
7   for  $j \in [0, k_b)$  do
8      $q_{i,j}[0..\ell-1] \leftarrow q_{i,j}[0..\ell-1] \oplus \delta_j \cdot d$  // perform
       $q[j] := q[j] + d[j] \cdot \Delta$  column-wise
9   for  $i \in [0..\ell + \lambda)$  do
10     $q[i] \leftarrow \text{ToField}(Q[i]; \lambda)$  // create VOLE keys row-wise
      (transpose  $Q$ )
11  $b \leftarrow 0$ ;
12  $((\alpha_1, \beta_1, \gamma_1), (\alpha_2, \beta_2, \gamma_2), q_{x_2}) \leftarrow \text{LinearTransform}(q_{pt}, q_{x_1}, \text{iv}_{\text{owf}}, ct)$ 
13  $\text{dot}_0 = \alpha_0^{\text{dot}} * \beta_1$   $\text{dot}_1 = \alpha_1^{\text{dot}} * \beta_2$ 
14  $b \leftarrow (\alpha_1.\text{mac} * \beta_1.\text{mac} - \gamma_1.\text{mac} * \Delta, \alpha_2.\text{mac} * \beta_2.\text{mac} - \gamma_2.\text{mac} * \Delta, \text{dot}_0 * \Delta + \Delta^2, \text{dot}_1 * \Delta + \Delta^2);$ 
15  $q^* \leftarrow \sum_{i \in [0..\lambda)} q[\ell + i] \alpha_\lambda^i$  //  $\alpha_\lambda \in \mathbb{F}_{2^\lambda}$ 
16  $\tilde{q} \leftarrow \text{ZKHash}(\text{chall}_2, (b \| q^*))$ 
17 return  $\tilde{q} - \tilde{a} \cdot \Delta$ 

```

2.3.4 Additional Functions.

Our construction relies on the same symmetric primitives as FAEST v1. Specifically, we use the following primitives:

- PRG : $\{0, 1\}^\lambda \times \{0, 1\}^{128} \rightarrow \{0, 1\}^*$, a pseudorandom generator taking as input a λ -bit seed and a 128-bit initialization vector.
- $H_0 : \{0, 1\}^{\lambda+128} \rightarrow \{0, 1\}^\lambda \times \{0, 1\}^{2\lambda}$, a hash function for commitment generation.
- $H_1 : \{0, 1\}^* \rightarrow \{0, 1\}^{2\lambda}$, a collision-resistant hash function.
- $H_2^j : \{0, 1\}^* \rightarrow \{0, 1\}^*$, for $j \in \{1, 2, 3\}$, hash functions used to derive the three Fiat–Shamir challenges and modeled as random oracles.
- $H_3 : \{0, 1\}^* \rightarrow \{0, 1\}^{\lambda+128}$, a hash function used to derive the signature randomness and initialization vector.

The pseudorandom generator PRG is instantiated using AES- λ in CTR mode with seed s and initialization vector iv_{vole} . We write $\text{PRG}(s, \text{iv}_{\text{vole}}; \ell)$ to denote the first ℓ output bits of the keystream.

All hash functions are instantiated identically to FAEST v1, using SHAKE128 for $\lambda = 128$ and SHAKE256 otherwise. To achieve domain separation, a one-byte domain identifier is appended to the input before hashing. For variable-length outputs, we write $H_i(x; \ell)$ to denote an ℓ -bit hash output.

Algorithm 16: ToField(x, k)

```

1 //  $\alpha_k$  is a primitive element of  $\mathbb{F}_{2^k}$ 
2 if  $x \in \{0, 1\}^k$  then
3   return  $\sum_{i=0}^{k-1} x[i] \alpha_k^i$ 
4 if  $x \in \{0, 1\}^{nk}$  then
5   for  $i \in [0..n)$  do
6      $y[i] := \sum_{j=0}^{k-1} x[ki + j] \alpha_k^j$ 
7   return  $y \in \mathbb{F}_{2^k}^n$ 
8 return  $\perp$ 

```

Algorithm 17: ToBits(x, k, n)

```

1  $y := \epsilon$ 
2 for  $i \in [0..n)$  do
3   Parse  $x[i] = x_0 + x_1 \alpha_k + \dots + x_{k-1} \alpha_k^{k-1}$ ,  $x_j \in \mathbb{F}_2$ 
4    $y := y \parallel (x_0 \dots x_{k-1})$ 
5 return  $y \in \{0, 1\}^{nk}$ 

```

Algorithm 18: ChalDec($\text{chall}_3, i; \text{param}$)

Input: $\text{chall}_3 \in \{0, 1\}^\lambda$, index $i \in [0..\tau)$, parameters param
Output: $(\delta_0, \dots, \delta_{k_i-1})$

```

1  $n_{\text{ch}} \leftarrow \lambda - w_{\text{grind}}$ 
2  $k_0 \leftarrow \lceil n_{\text{ch}} / \tau \rceil$ 
3  $k_1 \leftarrow \lfloor n_{\text{ch}} / \tau \rfloor$ 
4  $\tau_0 \leftarrow n_{\text{ch}} \bmod \tau$ 
5  $\tau_1 \leftarrow \tau - \tau_0$  //  $\tau_0 k_0 + \tau_1 k_1 = n_{\text{ch}}$ 
6 if  $i < \tau_0$  then
7    $k_i \leftarrow k_0$ 
8    $\text{off}_i \leftarrow i \cdot k_0$ 
9 else
10   $k_i \leftarrow k_1$ 
11   $\text{off}_i \leftarrow \tau_0 k_0 + (i - \tau_0) k_1$ 
12  $(\delta_0, \dots, \delta_{k_i-1}) \leftarrow \text{chall}_3[\text{off}_i : \text{off}_i + k_i]$ 
13 return  $(\delta_0, \dots, \delta_{k_i-1})$ 

```

Algorithm 19: DecodeAllChall3(chall₃; param)

```

1 for  $i \in [0..\tau)$  do
2    $(\delta_0, \dots, \delta_{k_i-1}) \leftarrow \text{ChalDec}(\text{chall}_3, i; \text{param})$ 
3    $I_i \leftarrow \sum_{j=0}^{k_i-1} \delta_j 2^j$  // same bits as ChalDec
4 return  $(I_0, \dots, I_{\tau-1})$ 

```

3 Security Goals

This section states the security goals of GreatWall and clarifies the scope of the security claims. The purpose of this section is not to provide the full proof, but to specify what is claimed by the scheme. The formal reductions are given in Sect. 6.6, and the dedicated cryptanalytic evaluation of Pylon is given in Sect. 6.

3.1 Correctness

GreatWall is required to satisfy correctness for honestly generated keys and honestly generated signatures. For a key pair (pt, ct) generated by the key-generation algorithm, where $ct = \text{Pylon}(\text{iv}_{\text{owf}}, pt)$, any signature σ honestly generated on a message m should be accepted by the verification algorithm, namely $\text{Verify}(ct, m, \sigma) = 1$. The signing algorithm may repeat the challenge generation when the grinding condition is not satisfied or when the batch all-but-one vector commitment opening aborts. These events only affect the expected signing time. Once the signing algorithm outputs a signature, the verifier deterministically checks the transcript and accepts an honestly generated signature except with negligible probability coming from the underlying randomized procedures.

3.2 Unforgeability

The primary security goal of GreatWall is existential unforgeability under adaptive chosen-message attacks. An adversary is given the public key and may adaptively request signatures on messages of its choice. The adversary wins if it outputs a valid signature on a new message that was not previously queried to the signing oracle.

GreatWall aims to achieve this security notion in both the classical random oracle model and the quantum random oracle model. In the classical setting, the adversary is a probabilistic polynomial-time algorithm with classical access to the random oracles. In the post-quantum setting, the adversary may be a quantum polynomial-time algorithm with quantum access to the random oracles, while signing queries remain classical.

At a high level, a successful forgery against GreatWall must either break the soundness of the VOLE-in-the-Head proof system or imply an algorithm for finding a preimage of the public Pylon output. The corresponding formal reductions are provided in Sect. 6.6.

3.3 One-Wayness of Pylon

The primitive-specific security goal of GreatWall is the preimage resistance of Pylon. Given a public value iv_{owf} and a target output ct generated as $ct = \text{Pylon}(iv_{\text{owf}}, pt)$ for a uniformly sampled secret input pt , it should be computationally infeasible to find any pt' such that $\text{Pylon}(iv_{\text{owf}}, pt') = ct$. This is the one-wayness assumption used by the GreatWall signature scheme, and it is the only primitive-specific assumption needed in the EUF-CMA reductions of Sect. 6.6.3 and Sect. 6.6.4.

This goal is analyzed from two complementary perspectives. First, Sect. 6.6.2 gives a generic preimage-security bound for Pylon in the ideal-permutation model, where the nonlinear layers are modeled as independent public random permutations. This analysis rules out generic black-box inversion strategies beyond the stated linear-query bound (Theorem 1), which requires $\Theta(2^n)$ queries to invert. Second, Sect. 6 evaluates the concrete Pylon instances against attacks that exploit their actual algebraic structure. The dedicated analysis includes brute-force preimage search in Sect. 6.1, algebraic attacks in Sect. 6.2, statistical attacks in Sect. 6.3, and quantum attacks in Sect. 6.4, with the overall conclusion summarized in Sect. 6.5.

Thus, the role of this goal is to connect the primitive-level security of Pylon with the signature-level security of GreatWall. Sect. 6.6 shows how a successful forgery can be reduced to finding a Pylon preimage, while Sect. 6 supports the claim that no known structural attack finds such a preimage below the claimed security level for the recommended parameters.

3.4 Scope of the Security Claims

The security claims of GreatWall apply to the algorithms and parameter sets specified in this document. The construction does not rely on number-theoretic, lattice-based, code-based, or multivariate public-key assumptions. Instead, its security is based on the preimage resistance of Pylon, the soundness and zero-knowledge properties of the VOLEitH proof system, and the random-oracle modeling of the hash functions used in the Fiat–Shamir transform.

The claims in this document are cryptographic security claims at the algorithmic level. Physical attacks, side-channel attacks, fault attacks, randomness failures, and attacks caused by incorrect or non-constant-time implementations are outside the scope of the formal security analysis and require separate implementation-level countermeasures.

GreatWall is specified at four security levels. Three variants—GreatWall-192, GreatWall-256, and GreatWall-512—correspond directly to the mandatory security levels of the NGCC solicitation. GreatWall-128 is provided as an additional variant for applications that do not require quantum security above 64 bits. Table 3 lists the correspondence between GreatWall variants and classical and quantum security strengths.

GreatWall’s security rests on two independent pillars: the one-wayness of the Pylon one-way function and the knowledge soundness of the underlying

Table 3: GreatWall variants and security levels.

Variant	n	NGCC Level	Classical security (bits)	Quantum security (bits)
GreatWall-128	137	Level 1	128	68
GreatWall-192	197	Level 1	192	80
GreatWall-256	263	Level 3	256	128
GreatWall-512	521	Level 5	512	256

VOLEitH proof system. An adversary who succeeds in forging a signature must either find a preimage of Pylon, or break the knowledge-soundness of the zero-knowledge argument; the precise reduction is given in Theorem 2. No structured algebraic hardness assumption is introduced at any point in the construction. The only complexity-theoretic assumptions made are the one-wayness of Pylon in the random permutation model and the existence of a hash function family that can be modelled as a (quantum) random oracle.

In order to fulfill the security claims stated in Table 3, implementations must ensure that each key pair is used with a fresh randomness seed per signing invocation and that the public key is transmitted and stored with integrity.

4 Features

4.1 A Modular Signature Design Centered on Pylon

GreatWall is organized as a modular signature scheme centered on the one-way function Pylon. The complete signature scheme, denoted by GreatWall, consists of the key-generation, signing, and verification algorithms, together with the VOLE-in-the-Head proof system, Fiat–Shamir challenges, GGM-tree-based batch all-but-one vector commitment, and the concrete parameter choices for each security level. In contrast, Pylon is the underlying one-way function that defines the public-key relation to be proven. The secret key is a preimage pt , and the public key is the corresponding output (iv_{owf}, ct) satisfying $ct = \text{Pylon}(iv_{\text{owf}}, pt)$.

This separation makes the role of each component explicit. The VOLEitH layer provides the generic proof-and-signature mechanism: it turns a proof of knowledge of a valid preimage into a publicly verifiable signature. Pylon, on the other hand, determines the concrete algebraic relation, the witness representation, and the main cost of the proof. Thus, GreatWall should be viewed as the full signature construction, while Pylon should be viewed as the dedicated one-way-function instance embedded inside that construction.

This modular structure is useful both for specification and for security analysis. The generic security of the signature scheme is analyzed through the soundness and zero-knowledge properties of the VOLEitH proof system, together with the Fiat–Shamir transform in the random oracle model and quantum random oracle model. The concrete security of the public-key relation is analyzed separately through the preimage resistance of Pylon. As a result, the overall security

claim of GreatWall is decomposed into a proof-system part and a primitive-specific part, which makes the assumptions and the scope of each analysis clear.

4.2 Proof-Cost-Aware One-Way Function Design

A distinguishing feature of GreatWall is that the underlying one-way function is designed with the cost structure of VOLEitH proofs in mind. In a VOLE-based proof, linear operations can be processed almost for free by exploiting the linearity of the VOLE correlations, whereas nonlinear multiplication constraints dominate the proof size, proving time, and verification cost. Therefore, the main design objective for Pylon is to keep the number of nonlinear constraints small without making the public relation algebraically simple.

This objective is achieved by using inverse Mersenne mappings as the only nonlinear components. For an inverse Mersenne S-box, the relation between its input x and output y can be checked by $xy = y^{2^e}$. The map $y \mapsto y^{2^e}$ is a Frobenius map and is therefore $\mathbb{F}_2$ -linear over $\mathbb{F}_{2^n}$. Hence the above relation contains only one genuine nonlinear operation, namely the multiplication xy . As a result, each S-box layer contributes only one field multiplication constraint to the QuickSilver-style consistency check.

This property has a direct impact on the proof relation used in GreatWall. The verifier does not need a separate nonlinear check for every linear layer, constant addition, or feed-forward addition. These parts are absorbed into linear transformations of the authenticated witness values. Consequently, the nonlinearity of the whole Pylon relation is concentrated in two multiplication checks, corresponding to the two inverse Mersenne layers.

This proof-cost-aware design is important for the overall efficiency of GreatWall. It reduces the extended witness length and the number of multiplication constraints, which are the dominant terms in VOLEitH-based signatures. At the same time, the low proof cost is not obtained by using a trivially invertible relation: the dense linear layers and nested feed-forward structure are included to preserve algebraic coupling between the two nonlinear layers. Thus, Pylon is optimized not merely as a standalone primitive, but as a one-way function whose algebraic form is suitable for efficient zero-knowledge proof generation.

4.3 Low-Round Structure with Security-Oriented Coupling

Pylon adopts a two-round structure in order to reduce the cost of proving the underlying one-way-function relation. In GreatWall, the size of the extended witness is directly affected by the number of full-state nonlinear layers. For an r -round construction of this form, the extended witness length is $\ell_r = rn$. Thus, replacing a three-round relation by a two-round relation removes one full n -bit nonlinear state from the witness. Since the witness is authenticated across all VOLE repetitions, this reduction is amplified in the final signature size.

However, the design does not rely on a low round number alone. A naive two-round construction may allow the internal relation to be separated into simpler one-round components, or may permit the surrounding linear layers to be merged

and eliminated by algebraic manipulation. Pylon therefore combines the two-round structure with dense linear layers and nested feed-forward connections. These components are introduced to preserve strong interaction between the two nonlinear layers even though the number of rounds is small.

The outer feed-forward branch binds the secret input to the final public output and prevents the two S-box layers from being treated as independent local transformations. The inner feed-forward branches inject earlier state information into later nonlinear computations, which increases the coupling among the input, intermediate values, and output. As a result, the public relation remains compact for the prover, but it is not reduced to a simple cascade of independently invertible layers.

This feature is essential for the design philosophy of GreatWall. The two-round architecture provides the main efficiency gain, while the security-oriented coupling mechanisms are used to compensate for the reduced round count. In this sense, Pylon is not merely a round-reduced primitive; it is a proof-efficient two-round one-way function whose internal dependencies are deliberately arranged to resist algebraic decomposition.

4.4 Parameter Selection Against Known Algebraic Shortcuts

The parameter choices of Pylon are driven not only by efficiency, but also by resistance against known algebraic shortcuts on power mappings over binary extension fields. In particular, the extension degrees are chosen to be prime, and the inverse Mersenne exponents are selected after checking the subgroup-linearization conditions that appear in recent attacks on round-reduced symmetric primitives.

This is important because power mappings over $\mathbb{F}_{2^n}$ may admit decompositions when $2^n - 1$ has suitable nontrivial divisors. In such cases, part of the nonlinear layer can sometimes be transformed into an $\mathbb{F}_2$ -linear Frobenius map after guessing a subgroup value. This may substantially reduce the effective number of variables and lead to guess-and-linearize or overdefined quadratic attacks. Pylon avoids this attack path by choosing prime extension degrees and by excluding exponents that admit such decompositions.

For the recommended instances, the exponents are screened so that the inverse Mersenne mappings cannot be rewritten in the form

$$x^{\bar{e}} = (x^d)^s \cdot x^{2^t}$$

for a useful nontrivial divisor $d \mid 2^n - 1$. Hence the attacker cannot turn an S-box layer into a linear residual map by a low-cost subgroup guess. The same screening is applied to both nonlinear layers, and the exponent pairs are chosen so that no common divisor enables simultaneous linearization of the two S-boxes.

This parameter-selection strategy is a feature of the construction rather than a post-hoc check. It ensures that the efficiency gained from using binary-field Frobenius structure does not introduce the subgroup structures required by known linearization-based attacks. The concrete divisor checks and dedicated

algebraic analysis are given later in the security evaluation, while this section highlights the design principle behind the selected parameters.

4.5 Scalable Security Levels with a Uniform Architecture

GreatWall is designed to scale across different security levels without changing the overall structure of the signature scheme or the underlying one-way function. The scheme provides four parameter sets, namely GreatWall-128, GreatWall-192, GreatWall-256, and GreatWall-512. These instances use the same Pylon architecture, the same VOLEitH proof framework, and the same signing and verification procedures. Across security levels, the main changes are the field size, the inverse Mersenne exponents, and the VOLEitH parameters required to meet the corresponding security target.

This uniformity is useful for both implementation and analysis. From the implementation perspective, the same code structure can be reused across all four security levels, with only parameter-dependent field arithmetic and proof parameters being adjusted. From the analysis perspective, the same security arguments apply to all instances: the generic signature security is derived from the VOLEitH proof system and the Fiat–Shamir transform, while the concrete one-wayness analysis is instantiated with the corresponding Pylon parameters.

The scalability of Pylon mainly comes from enlarging the binary extension field. Increasing n directly increases the preimage-search space and the cost of generic algebraic attacks, while preserving the same two-round architecture and the same type of S-box relation. Hence higher-security instances do not require adding more rounds, introducing additional parallel branches, or replacing the primitive with a different construction.

This feature is particularly important for long-term post-quantum deployment. GreatWall provides a consistent path from 128-bit security to conservative long-term security levels, while keeping the specification, implementation framework, and security analysis uniform across the family.

4.6 Compact-Key and Signature-Size-Oriented Performance

GreatWall is designed with compact keys and signature-size-oriented performance in mind. Since the public key is one output of the Pylon one-way function and the secret key is the corresponding preimage, the key sizes remain small across all parameter sets. This feature is useful in applications where public keys, certificates, or authentication data need to be transmitted, stored, or verified frequently.

The dominant communication cost of GreatWall comes from the VOLEitH proof rather than from the public-key material. Therefore, the performance-oriented part of the design focuses on reducing the size of the relation that must be proven. Pylon contributes to this goal in two ways. First, its two-round structure reduces the extended witness length by removing one full nonlinear state compared with a three-round construction. Second, its inverse Mersenne S-boxes

keep the number of multiplication constraints small, so that the QuickSilver-style consistency check remains compact.

This gives GreatWall a clear performance profile within the family of symmetric-key-based and VOLEitH-based signatures. The scheme combines compact public keys with a proof relation specifically optimized for short VOLEitH signatures. The resulting construction is intended to provide a practical balance between signature size, signing time, and verification time on standard 64-bit platforms.

Overall, the performance advantage of GreatWall comes from the co-design of the signature framework and the underlying one-way function. The VOLEitH layer supplies a modular proof-and-signature mechanism, while Pylon reduces the cost of the proven relation. This combination allows GreatWall to obtain shorter signatures among comparable VOLEitH-based designs while preserving a simple and scalable construction, without relying on number-theoretic, lattice-based, code-based, or multivariate public-key assumptions.

5 Design Rationale

5.1 Choice of a Two-Round Architecture.

The number of rounds of the underlying one-way function affects the VOLEitH signature size through the extended witness length ℓ . In our setting, each full-state nonlinear layer contributes one state of n authenticated bits to the proof relation, while the surrounding linear layers, constants, and feed-forward operations are handled locally by the linearity of VOLE. Hence, for an r -round construction of this form, the extended witness length is $\ell_r = rn$. This observation motivates minimizing the number of nonlinear rounds. A three-round construction gives $\ell_3 = 3n$, whereas a two-round construction gives $\ell_2 = 2n$. Therefore, moving from three rounds to two rounds reduces the extended witness length by $\Delta\ell = \ell_3 - \ell_2 = n$. Thus this translates into an absolute signature-size saving of

$$\Delta|\sigma| = \tau n \tag{4}$$

bits for all VOLEitH-based instantiations considered in this paper.

However, achieving security at two rounds is far from automatic. The recent break of two-round Rain [44] demonstrates that naively removing nonlinear rounds may substantially weaken the resulting algebraic system. Instead, Pylon addresses this problem by combining two full-state inverse Mersenne S-box layers with dense linear layers and a *nested feed-forward structure*. The dense linear layers spread each nonlinear transformation over the entire state, while the feed-forward terms make the two nonlinear layers couple more tightly and prevent a direct separation into two independent one-round problems.

Together, these mechanisms are intended to compensate for the reduced number of rounds and to preserve sufficient algebraic complexity in the resulting one-way function. And concrete signature-size reductions obtained from this choice are reported in Sect. 7.

5.2 Choice of Field.

For a symmetric primitive intended for VOLEitH-style zero-knowledge protocols, the choice of the underlying field must balance efficiency in secure evaluation and resistance against algebraic cryptanalysis. Binary extension fields $\mathbb{F}_{2^n}$ are naturally attractive in this setting, since Frobenius maps $x \mapsto x^{2^t}$ are $\mathbb{F}_2$ -linear and can be implemented very efficiently. However, this same structure may also introduce additional algebraic regularity that can be exploited by linearization-based attacks. In particular, recent attacks on round-reduced RAIN and full AIM-III [44] show that, when $2^n - 1$ admits suitable nontrivial divisors, a power mapping can sometimes be rewritten in the form $x^k = (x^d)^s \cdot x^{2^t}$, so that after guessing the subgroup value x^d , the remaining part becomes an $\mathbb{F}_2$ -linear map and the nonlinear layer can be partially reduced to a linear or low-degree system.

To avoid this structural weakness, we choose the extension degree n to be a prime number. This choice is motivated by security rather than implementation convenience. When n is prime, the factorization structure of $2^n - 1$ is much more restrictive than in the even-degree case, and in particular it excludes the special decompositions exploited in the known linearization attacks on inverse- and Mersenne-type power maps. For our target application in VOLEitH, this is especially desirable: the primitive is used as a low-multiplicative-complexity one-way function inside a VOLE-in-the-head framework, where any algebraic shortcut directly threatens the soundness margin of the proof system. By working over $\mathbb{F}_{2^n}$ with prime extension degree, we retain the efficiency benefits of binary-field arithmetic while deliberately avoiding extension degrees that admit the subgroup structures required by such attacks. Therefore, the choice of field in our design is guided not only by implementation efficiency, but also by the need to suppress linearization-based algebraic attacks in the VOLEitH setting.

We note that the chosen prime degrees n are only slightly larger than the security parameter λ , giving an overhead at most 7% in field-element size and less than 3% for $\lambda \geq 192$. This marginal gap has a negligible impact on arithmetic cost. The extended witness length increases from 2λ to $2n$ bits, contributing fewer than 18 additional authenticated bits per repetition at the 128-bit level, which is absorbed into the constant terms of the signature-size formula. In practice, the benchmarks in Section 7 confirm that this overhead is not a bottleneck.

5.3 Choice of Nonlinear Layer: Inverse Mersenne Mapping.

In the VOLEitH setting, the proof cost is closely tied to the nonlinear part of the underlying relation. Linear operations can usually be handled very efficiently, whereas each nonlinear multiplication contributes directly to the online proving cost and also affects the overall proof size and verification complexity. Therefore, the nonlinear layer should satisfy two requirements simultaneously: it should be economical to represent inside the VOLEitH protocol, and it should avoid algebraic structures that admit overly compact equation systems. To achieve this goal, we instantiate the nonlinear layer by the *inverse Mersenne mapping*

$$S(x) = x^{\bar{e}}, \quad \bar{e} = (2^e - 1)^{-1} \pmod{2^n - 1}, \quad (5)$$

over $\mathbb{F}_{2^n}$, where $\gcd(e, n) = 1$ so that S is a permutation. This choice is motivated by the following considerations.

First, the inverse Mersenne mapping provides a flexible family of power permutations with high algebraic degree, while being more tunable than the fixed inverse mapping $x^{-1} = x^{2^n-2}$. In a VOLEitH protocol, the nonlinear layer is not used in isolation, but as part of an efficiently provable one-way-function relation. Hence, it is desirable to have a family of exponents that allows the proof cost and the algebraic properties of the primitive to be optimized jointly. The inverse Mersenne form offers exactly this flexibility: by varying the parameter e , one can search for exponents that retain strong algebraic complexity while leading to more efficient circuit representations than plain inversion.

Second, this choice is attractive in the VOLEitH setting because it admits a simple algebraic verification rule. Let

$$y = x^{\bar{e}}, \quad \bar{e} = (2^e - 1)^{-1} \pmod{2^n - 1}. \quad (6)$$

Then one equivalently has

$$y^{2^e-1} = x, \quad \text{or equivalently} \quad y \cdot x = y^{2^e}. \quad (7)$$

Hence, the S-box can be verified through a single multiplication constraint, which is particularly efficient in VOLEitH. This gives the inverse Mersenne mapping a favorable feature as a nonlinear layer with both low proof cost and strong algebraic structure.

Third, from the perspective of algebraic modeling, this choice also offers an important security advantage. A standard inverse S-box admits relatively many implicit quadratic relations between its input and output, and in particular it is known to yield up to $5n$ linearly independent quadratic equations. By contrast, Mersenne-type power mappings yield only $3n$ such equations. Therefore, using an inverse Mersenne mapping helps limit the number of low-degree relations available to the attacker, making it harder to derive an overdefined quadratic system [29]. This is particularly desirable in the VOLEitH setting: while the proof system benefits from low multiplicative complexity, the primitive should not at the same time expose too many quadratic constraints that could simplify algebraic cryptanalysis.

5.3.1 Choice of the Exponents.

The exponents e_1 and e_2 are selected according to both VOLEitH efficiency and algebraic-security requirements.

On the efficiency side, we require that the corresponding inverse exponents

$$\bar{e}_i = (2^{e_i} - 1)^{-1} \pmod{2^n - 1} \quad (8)$$

admit efficient arithmetic realizations and lead to low nonlinear cost inside the VOLEitH proof system. Since the proof cost is dominated by nonlinear operations, the exponents must be chosen so that the resulting S-boxes remain

practical to evaluate while still providing sufficiently strong algebraic mixing. In particular, we favor exponents whose inverse Mersenne forms can be implemented with short multiplication chains, so that both prover computation and multiplication verification remain efficient.

On the algebraic side, we require $\gcd(e_i, n) = 1$ so that $x \mapsto x^{2^{e_i}-1}$, and hence also its inverse $x \mapsto x^{\bar{e}_i}$, is a permutation over $\mathbb{F}_{2^n}$. Moreover, the exponents are chosen so that the induced inverse Mersenne mappings have high algebraic degree and ensure rapid degree growth across rounds, which is important for reaching full-state algebraic diffusion with only a small number of rounds. At the same time, we avoid exponents that would introduce excessive low-degree structure or unusually sparse quadratic descriptions.

Most importantly, the inverse Mersenne exponents $\bar{e}_1, \bar{e}_2$ are selected to avoid known linearization-based algebraic attacks. Concretely, for our parameter sets there is no nontrivial divisor $d \mid (2^n - 1)$ such that $\bar{e}_i \equiv 2^t \pmod{d}$ for some integer t and for either $i \in \{1, 2\}$. Hence the nonlinear mapping $S_i(x) = x^{\bar{e}_i}$ cannot be rewritten in the form $x^{\bar{e}_i} = (x^d)^s \cdot x^{2^t}$ with a nontrivial subgroup component and an $\mathbb{F}_2$ -linear residual term. In addition, the pair (e_1, e_2) is chosen so that no common nontrivial divisor $d \mid (2^n - 1)$ simultaneously enables such a decomposition for both nonlinear layers. Therefore, the simultaneous linearization mechanism underlying known attacks on power mappings over $\mathbb{F}_{2^n}$ does not apply to our inverse Mersenne S-boxes [44].

5.3.2 Linear Layer: Random Dense Matrix.

For the linear layer, we use multiplication by a randomly chosen dense matrix $M \in \mathbb{F}_2^{n \times n}$. The matrices are selected so that their linearized-polynomial representations are dense and have max degree of 2^{n-1} in $\mathbb{F}_{2^n}$. This unstructured affine layer serves two critical purposes. First, it ensures full diffusion at the bit level. Second, when combined with the nonlinear operations and feed-forward structure, it rapidly increases the degree of regularity of the algebraic system after only a few rounds, making algebraic attacks computationally infeasible.

5.3.3 Nested Feed-forward Structure.

Pylon employs three forward-feedback connections to enhance cross-round interaction and to obstruct algebraic decomposition. Instead of letting each round be viewed as an almost independent composition of one linear layer and one S-box, the feedback paths repeatedly re-inject earlier state information into later stages, so that intermediate variables become tightly fused throughout the entire permutation. This is especially desirable in our design, since recent decomposition-based algebraic attacks on arithmetic-oriented primitives [39] explicitly exploit structural simplifications of the internal relation, either by splitting it into simpler low-degree components or by reducing several linear layers to a single equivalent one. The forward-feedback structure is therefore introduced to strengthen the coupling among intermediate variables and to obstruct both types of simplification. A quantitative assessment of this security contribution is given in

Sect. 6.2, where we analyze two stripped variants of Pylon with feed-forward connections partially or fully removed: without any feed-forward the function becomes directly invertible at cost $O(n^\omega)$, and retaining only the outer feed-forward reduces the attack complexity to approximately $2^{n-1+e_1+e_2}$, far below the $2^{\omega(n-1+e_1+e_2)}$ cost of the full construction. This confirms that both layers of feed-forward are necessary for the two-round design to achieve the target security level.

6 Security Evaluations

6.1 Brute-force Attack

A straightforward way to attack Pylon is exhaustive search over the whole preimage space. Given a public instance (iv_{owf}, ct) , where $ct = \text{Pylon}(iv_{\text{owf}}, pt)$ for an unknown secret input pt , an adversary may simply enumerate all 2^n possible candidates $pt \in \mathbb{F}_2^n$, evaluate $\text{Pylon}(iv_{\text{owf}}, pt)$, and check whether the result equals ct . Therefore, the time complexity of the brute-force preimage attack is $O(2^n)$, with average complexity about 2^{n-1} , while the memory is negligible.

6.2 Algebraic Attacks

In this part, we analyze algebraic attacks on Pylon. It must be emphasized that Pylon is used as a one-way function within the VOLEitH framework. To be specific, the adversary is restricted to a known pair (iv_{owf}, ct) , aimed at recovering the secret input pt , s.t., $\text{Pylon}(iv_{\text{owf}}, pt) = ct$.

Gröbner Basis Attacks. Our primary aim is to model Pylon as a system of algebraic equations. To solve for the unknown variables, we use Gröbner basis algorithms [16]. By performing an order transformation on the computed basis, the multivariate system can be converted into a triangular form containing univariate equations, which are then solvable. The system of equations can be modeled by several ways.

Equation System over $\mathbb{F}_2$. We analyze Gröbner basis attacks by modeling Pylon as a Boolean polynomial system over $\mathbb{F}_2$. We introduce intermediate state variables between rounds so that each round contributes a block of quadratic constraints induced by the nonlinear layer. With r rounds, this gives a quadratic system with rn Boolean variables and vrn quadratic equations, together with the standard Boolean field equations $x^2 - x = 0$ for all variables. Under the semi-regular assumption [3], the corresponding Hilbert series can be written as $HS(z) = (1+z)^{rn}(1-z^2)^{vrn}$. We estimate the degree of regularity d_{reg} as the smallest degree d at which the coefficient of z^d in $HS(z)$ becomes non-positive. Then the complexity of computing a Gröbner basis can be estimated by the standard linear-algebra bound $O\left(\binom{rn+d_{\text{reg}}}{d_{\text{reg}}}\omega\right)$, where ω is usually set to 2. In our concrete parameters, we set $r = 2$ and $v = 3$ since the inverse Mersenne

mapping provides $3n$ linearly independent quadratic equations over $\mathbb{F}_2$ on the n state bits, hence

$$\text{HS}(z) = (1 + z)^{2n}(1 - z^2)^{6n}. \quad (9)$$

The resulting d_{reg} estimates and the corresponding bit complexities are summarized in Table 4.

Scheme	n	v	d_{reg}	Bit complexity
Pylon-128	137	3	15	163.7
Pylon-192	197	3	20	224.2
Pylon-256	263	3	25	286.3
Pylon-512	521	3	45	532.5

Table 4: Estimated degree of regularity and complexity of Gröbner-basis attacks over $\mathbb{F}_2$ for Pylon.

Experimental Validation on Reduced Instances. We performed concrete Gröbner-basis computations on reduced Pylon instances over $\mathbb{F}_2$. For each reduced dimension n , we generated random inputs $x \in \mathbb{F}_{2^n}$, computed $y = \text{Pylon}(\text{iv}_{\text{owf}}, x)$, and encoded the inversion problem $\text{Pylon}(\text{iv}_{\text{owf}}, X) = y$ as a Boolean polynomial system. We used the same intermediate-state formulation as in the complexity estimate, with $r = 2$ rounds and $v = 3$ quadratic constraints per state bit.

We stress that this formulation is favorable to the attacker, since the use of intermediate variables keeps the system quadratic and avoids the high-degree expansion of the composed Pylon map. The Gröbner bases were computed in grevlex order using Magma [13]. We recorded the solving degree sd , defined as the highest degree reached during the Gröbner-basis computation.

The results are shown in Table 5. For the solved instances ($n \leq 19$), the observed solving degree matches the Hilbert-series estimate ($sd = d_{\text{reg}}$ or $d_{\text{reg}} - 1$), consistent with the semi-regularity assumption [3]. The running time grows rapidly with n : instances with $n \geq 23$ did not terminate within the time limit.

Extended Degree-Limited Experiments. To further probe the behavior at larger dimensions, we ran an additional set of degree-limited Gröbner-basis computations for $n \in \{23, 29, 31, 37\}$ using Magma’s `DegreeLimit` parameter set to d_{reg} . In all four cases, the computation reached step degree 4 but did not advance to the predicted $d_{\text{reg}} \in \{5, 6\}$ within the 300s wall-clock limit (Table 6). Peak memory consumption (Peak RSS) grew from 1.34 GB at $n = 23$ to 10.33 GB at $n = 37$, indicating exponential growth of intermediate basis elements as the computation approaches d_{reg} . This is consistent with semi-regular system behavior.

Crucially, *no anomalous low-degree termination was observed in any instance*. The basis computation progressed normally through degree 4 without early collapse or unexpected degree drop. Early termination below the predicted d_{reg} would be the signature of an exploitable low-degree structure; its absence across all tested dimensions confirms that Pylon does not exhibit such weakness. The inability to reach d_{reg} within the resource limit is consistent with the exponen-

tial intermediate-basis growth characteristic of semi-regular systems, rather than any structural shortcut.

n	r	v	#Var.	#Quad.	Eq.	d_{reg}	sd	Median time	Result
11	2	3	22	66		4	4	0.10 s	5/5 solved
13	2	3	26	78		4	4	0.90 s	5/5 solved
17	2	3	34	102		4	4	21.37 s	5/5 solved
19	2	3	38	114		5	4	98.06 s	5/5 solved
23	2	3	46	138		5	–	> 3 h	timeout
29	2	3	58	174		6	–	> 3 h	timeout

Table 5: Experimental Gröbner-basis attacks on reduced Pylon instances over $\mathbb{F}_2$. For $n \leq 19$, we generated five random instances and report the median solving time. For $n \in \{23, 29\}$, the first random instance did not terminate within the time limit. The sd denotes the highest degree reached during the grevlex Gröbner-basis computation.

n	d_{reg}	DegreeLimit	Step	degree reached	Peak RSS	Result
23	5	5		4	1.34 GB	timeout (> 300 s)
29	6	6		4	5.36 GB	timeout (> 300 s)
31	6	6		4	5.11 GB	timeout (> 300 s)
37	6	6		4	10.33 GB	timeout (> 300 s)

Table 6: Degree-limited Gröbner-basis experiments for larger reduced Pylon instances. **DegreeLimit** is set to d_{reg} in each case. The step degree reached before timeout is 4 in all instances, with no anomalous low-degree termination.

Equation System over $\mathbb{F}_{2^n}$. Another approach is to construct the equation systems over $\mathbb{F}_{2^n}$. We can model the function as a univariate polynomial equation. However, due to the high algebraic degree of the inverse Mersenne S-box and the dense linear layer, the resulting polynomial reaches nearly the maximal degree of $(2^n - 2)$ after a single round. Root-finding algorithms for such high-degree dense polynomials require a complexity of approximately $\mathcal{O}(2^{3n})$ [21], rendering this approach computationally intractable. Consequently, we focus our security analysis on multivariate equation systems over $\mathbb{F}_{2^n}$. Considering the fact that $y = x^{\bar{e}} \implies x = y^{2^{\bar{e}}-1}$, we can add intermediate variables to decrease the degree of the inverse Mersenne S-box. By introducing an intermediate variable x_i in each round i after S-box, we obtain equation systems in the following:

$$\begin{cases} M_1(pt) + \gamma_1 = x_1^{2^{e_1}-1}, \\ M_2(x_1 + pt) + \text{iv}_{\text{owf}} = x_2^{2^{e_2}-1}, \\ M_3(x_2) + x_2^{2^{e_2}-1} + \gamma_2 + pt = ct. \end{cases} \quad (10)$$

Here we denote the S-box output in round i by x_i . Note that the three equations have a high degree, since M_i is a linearized polynomial of maximum high degree

of 2^{n-1} in our construction. Hence the degree of regularity [2] in this system can be assumed to be $D_{reg} = 1 + \sum_{i=1}^3 \deg(f_i) - 1 \geq 2^{n-1}$, hence we can see that

$$\binom{r+1+2^{n-1}}{2^{n-1}}^\omega > \binom{1+2^{n-1}}{2^{n-1}}^\omega = (1+2^{n-1})^\omega > 2^n, \quad (11)$$

namely $\log_2 \left(\binom{r+1+2^{n-1}}{2^{n-1}}^\omega \right) > n$. So it is infeasible for Gröbner Basis attacks.

Overdefined Quadratic Modeling Attacks. A recent line of work demonstrated that 3-round RAIN can be modeled as a highly overdefined multivariate quadratic polynomial system by exploiting the field inversion at the boundary rounds [29]. More specifically, their construction uses identities that are special to $y = x^{-1}$, following the idea of Courtois–Pieprzyk [15].

Let x_1 denote the output of the first inversion layer and x_2 the input to the last inversion layer in 3-round RAIN, and let $\vec{x}_1, \vec{x}_2 \in \mathbb{F}_2^\ell$ be the coordinate (bit) vectors of $x_1, x_2 \in \mathbb{F}_{2^\ell}$ with respect to a fixed basis of $\mathbb{F}_{2^\ell}$. This allows them to derive an extra 5ℓ quadratic Boolean equations that involve only $(\vec{x}_1, \vec{x}_2)$, which makes the resulting system significantly more overdefined. This mechanism is closely linked to the algebra of the inverse function. Besides the basic relation $xy = 1$, one can derive up to $5n$ linearly independent quadratic Boolean relations from $y = x^{-1}$ when it is expanded over $\mathbb{F}_2$. Our design deliberately replaces the inversion with the inverse Mersenne S-box $S(x) = x^e$. In contrast to the inverse map, the inverse Mersenne S-box admits only about $3n$ linearly independent quadratic Boolean relations of the type used in quadratic modeling, as analyzed in AIM2 [25]. Therefore, the specific overdefinition boost exploited for 3-round RAIN does not directly carry over to our setting, and the attacker cannot obtain the same amount of free quadratic constraints from the S-box layer alone. In addition, our surrounding linear layers and the feed-forward structure further hinder the boundary elimination tricks that are essential in [29]. Taken together, we do not expect the multivariate quadratic system from our construction to become highly overdefined, which is required for this type of attack. Therefore, we consider this attack ineffective for our chosen parameters.

Resultant Attacks. Resultant-based algebraic attacks have recently attracted increasing attention in the cryptanalysis of symmetric primitives. In particular, such attacks have been successfully applied to several arithmetic (AO) primitives, including Rescue-Prime [40,4], JARVIS [40], AIM [39], AIM2 [34] and so on. These results demonstrate that even primitives designed for algebraic proof systems can still have algebraic weaknesses when their internal structure produces systems with usable resultants. In this part, we analyze the resistance of Pylon against resultant attacks. We follow the general attack framework proposed in prior attacks on AIM [39] and AIM2 [34].

MITM Modeling for Pylon. We apply a meet-in-the-middle (MITM) modeling strategy to the algebraic system induced by our construction [39]. Specifically, we introduce M_a and subsequently introduce the inverse matrices M_a^{-1} in order to

cancel the effect. Let $L_1(x), L_2(x), L_3(x)$ and $L_a(x)$ denote the linearized polynomials associated with the matrices M_1, M_2, M_3 and M_a , respectively. Denote $L_\alpha(x) = L_a(L_1(x)), L_\beta(x) = L_a(L_2^{-1}(x)), L_\gamma(x) = L_a(L_3(x))$, and the output of the first two S-box as v_1 and v_2 . This modeling derives a system of bivariate equations as

$$\begin{cases} L_\alpha(pt) = L_a(v_1^{2^{e_1}-1}), \\ L_a(v_1 + pt) = L_\beta(v_2^{2^{e_2}-1}), \\ L_\gamma(v_2) = L_a(pt + ct + v_2^{2^{e_2}-1}). \end{cases} \quad (12)$$

To eliminate the variable v_1 and v_2 , we rewrite the equation as

$$\begin{cases} g_1 := L_\alpha(pt) + L_a(v_1^{2^{e_1}-1}) = 0, \\ g_2 := L_a(v_1 + pt) + L_\beta(v_2^{2^{e_2}-1}) = 0, \\ g_3 := L_\gamma(v_2) + L_a(pt + ct + v_2^{2^{e_2}-1}) = 0. \end{cases} \quad (13)$$

Constructing Low-degree Equations. Based on the linearized polynomial decomposition method [39], let

$$L_1(x) = \sum_{j=0}^{n-1} d_{1,j}x^{2^j}, L_2^{-1}(x) = \sum_{j=0}^{n-1} d_{2,j}x^{2^j}, L_3(x) = \sum_{j=0}^{n-1} d_{3,j}x^{2^j}, L_a(x) = \sum_{i=0}^{n-1} a_i x^{2^i},$$

where $d_{1,j}$'s, $d_{2,j}$'s, $d_{3,j}$'s are known coefficients while a_i 's are to be determined. Assume that

$$\deg(L_a(x)) \leq 2^l, \deg(L_\alpha(x)) \leq 2^\gamma, \deg(L_\beta(x)) \leq 2^\beta, \deg(L_\gamma(x)) \leq 2^\tau,$$

then we can get a system of $(3n - \gamma - \beta - \tau - 3)$ equations with $(l+1)$ unknowns:

$$\begin{cases} a_i = 0 & \text{for } l+1 \leq i \leq n-1, \\ \sum_{\substack{0 \leq i, j \leq n-1 \\ i+j \equiv t_1 \pmod n}} a_i d_{1,j}^{2^i} = 0 & \text{for } \gamma+1 \leq t_1 \leq n-1, \\ \sum_{\substack{0 \leq i, j \leq n-1 \\ i+j \equiv t_2 \pmod n}} a_i d_{2,j}^{2^i} = 0 & \text{for } \beta+1 \leq t_2 \leq n-1, \\ \sum_{\substack{0 \leq i, j \leq n-1 \\ i+j \equiv t_3 \pmod n}} a_i d_{3,j}^{2^i} = 0 & \text{for } \tau+1 \leq t_3 \leq n-1. \end{cases} \quad (14)$$

For a nontrivial solution to exist, the variable count should exceed the equation count, i.e., $l + \gamma + \beta + \tau + 4 > 3n$.

Extracting a Univariate Equation. The goal is to eliminate the variables v_1 and v_2 to obtain a univariate polynomial of pt . Note that the substitution theory [40] is not applicable here. Firstly we compute $G_1(pt, v_2) = R(g_1, g_2, v_1)$ to eliminate v_1 . The size of the Sylvester matrix is at most $(2^l + 2^l \cdot (2^{e_1} - 1)) \times (2^l + 2^l \cdot (2^{e_1} - 1))$, and after this the degree of pt and v_2 are bounded by $D_1 := \deg_{pt}(G_1) \leq 2^l \cdot 2^l \cdot (2^{e_1} - 1) + 2^\gamma \cdot 2^l$ and $\deg_{v_2}(G_1) \leq 2^\beta \cdot (2^{e_2} - 1) \cdot 2^l \cdot (2^{e_1} - 1) \approx 2^{\beta+l+e_1+e_2}$. Secondly we compute $G_2(pt) = R(G_1, g_3, v_2)$ to eliminate v_2 . The degree of v_2 in g_3 is $D_2 := \deg_{v_2}(g_3) = \max(2^\tau, (2^{e_2} - 1) \cdot 2^l)$. The Sylvester matrix has size

of most $(2^{\beta+l+e_1+e_2} + D_2) \times (2^{\beta+l+e_1+e_2} + D_2)$, and after this the degree of pt is bounded by

$$d := \deg_{pt}(G_2) \leq D_1 \cdot D_2 + 2^l \cdot 2^{\beta+l+e_1+e_2}. \quad (15)$$

Finally we solve $G_2(pt)$ and get the solution.

Complexity Analysis. The main computational cost stems from the resultant elimination of v_2 and solving $G_2(pt)$. The complexity of eliminating v_2 [5] can be estimated as

$$\mathcal{O}(D_1(\log D_1)^2(2^{\beta+l+e_1+e_2} + D_2)^\omega), \quad (16)$$

and the complexity of solving $G_2(pt)$ [6,36] can be estimated as

$$\mathcal{O}(d \log d (\log d + \log q \log \log d)). \quad (17)$$

Limitation of the MITM resultant approach. Although the above MITM-resultant strategy is well-defined, it does not lead to a competitive attack on Pylon. Unlike the case of AIM, the intermediate variables in Pylon cannot be merged into a single variable together with a useful substitution structure. As a result, one has to eliminate v_1 and v_2 sequentially. However, the first elimination already causes a significant blow-up in the degree with respect to the remaining variable v_2 , namely $\deg_{v_2}(G_1)$ becomes very large. Consequently, the second resultant dominates the overall cost, and the resulting complexity is higher than that of direct algebraic modeling. Hence, we do not regard the above resultant-based approach as the best attack on Pylon, and turn to direct modeling.

Direct Modeling. Let $L_1(x)$, $L_2(x)$ and $L_3(x)$ denote the linearized polynomials associated with the matrices M_1 , M_2 and M_3 , respectively. Denote the output of the second S-box as v . Then we have

$$\begin{cases} L_1(pt) + \gamma_1 = (L_2^{-1}(v^{2^{e_2}-1} + \text{iv}_{\text{owf}}) + pt)^{2^{e_1}-1}, \\ L_3(v) + v^{2^{e_2}-1} + \gamma_2 + pt = ct. \end{cases} \quad (18)$$

To eliminate the variable v , we rewrite the equation as

$$\begin{cases} g_1 := L_1(pt) + \gamma_1 + (L_2^{-1}(v^{2^{e_2}-1} + \text{iv}_{\text{owf}}) + pt)^{2^{e_1}-1} = 0, \\ g_2 := L_3(v) + v^{2^{e_2}-1} + \gamma_2 + pt + ct = 0. \end{cases} \quad (19)$$

We compute $G(pt) = R(g_1, g_2, v)$ to eliminate the variable v . The Sylvester matrix has size of most $(2^{n-1} \cdot (2^{e_2}-1) \cdot (2^{e_1}-1) + 2^{n-1}) \times (2^{n-1} \cdot (2^{e_2}-1) \cdot (2^{e_1}-1) + 2^{n-1})$, and after this we have $d := \deg_{pt}(G) \leq 2^{n-1} \cdot (2^{e_2}-1) \cdot (2^{e_1}-1) + 2^{n-1} \cdot 2^{n-1}$. Following the complexity formula in [39] introduced above, the dominant cost is

$$\log_2 T \approx \underbrace{(n-1) + 2 \log_2(n-1)}_{\log_2[D_2(\log D_2)^2]} + \underbrace{\omega(n-1+e_1+e_2)}_{\omega \log_2 D_1}, \quad (20)$$

where $D_1 = 2^{n-1}(2^{e_1}-1)(2^{e_2}-1)$ and $D_2 = 2^{n-1}$ are the degrees of g_1 and g_2 in v , respectively. We note that D_1 already exceeds the brute-force cost 2^n for all recommended parameter sets, so the resultant computation alone is more expensive than exhaustive search. In our parameter setting, the estimated complexity of the above direct modeling resultant attack is summarized in Table 7, which shows the security of Pylon against resultant attack.

Role of the Feed-Forward Structure. To quantify the security contribution of the nested feed-forward structure, we analyze two stripped variants of Pylon.

Case 1: No feed-forward. Without any feed-forward connection, M_1 and M_3 can be eliminated by direct substitution, reducing the system to

$$\begin{cases} g_1 := pt + t_1^{2^{e_1}-1} = 0, \\ g_2 := M_2(t_1) + iv_{\text{owf}} + ct^{2^{e_2}-1} = 0. \end{cases} \quad (21)$$

where $ct^{2^{e_2}-1}$ is a known constant. Since g_2 is linear in t_1 , the unique solution $t_1 = M_2^{-1}(ct^{2^{e_2}-1} - iv_{\text{owf}})$ is recovered by a single matrix inversion, and $pt = t_1^{2^{e_1}-1}$ follows immediately. The total attack cost is $O(n^\omega)$ field operations, which is at most 2^{26} for all recommended parameter sets—the function is directly invertible and provides no one-wayness guarantee.

Case 2: Outer feed-forward only. Retaining the outer feed-forward while removing the two inner connections, the relation $pt = ct + M_3(v) + \gamma_2$ allows pt to be substituted out, and $M_1 \circ M_3$ is absorbed into a single redefined linear map $\tilde{M} := M_1 \circ M_3$, so that the first and third linear layers introduce no additional degree. The system becomes

$$\begin{cases} g_1 := t_1^{2^{e_1}-1} + \tilde{M}(v) + c_1 = 0, \\ g_2 := M_2(t_1) + iv_{\text{owf}} + v^{2^{e_2}-1} = 0. \end{cases} \quad (22)$$

where c_1 is a known constant. Again g_2 is linear in t_1 , so t_1 is eliminated by direct substitution, yielding a univariate polynomial in v of degree $2^{n-1}(2^{e_1}-1)(2^{e_2}-1)$. The attack cost is dominated by solving this polynomial, with $\log_2 T \approx n - 1 + e_1 + e_2$, giving approximately 2^{281} for Pylon-128—above the security target but far below the 2^{937} cost of the full construction.

Taken together, these two cases show that the inner feed-forward connections are responsible for preventing the linear elimination of t_1 from g_2 : in the full Pylon, the inner paths inject pt into the input of the second S-box, raising $\deg_{pt}(g_1)$ from 1 to $2^{e_1} - 1$ and forcing a full resultant computation whose cost grows as $2^{\omega(n-1+e_1+e_2)}$. The outer feed-forward in turn prevents round decomposition by binding the two S-box layers to a single global input-output relation. Both layers of feed-forward are therefore necessary for the two-round design to achieve its target security level.

Scheme	n	Time complexity
Pylon-128	137	937.0
Pylon-192	197	1317.2
Pylon-256	263	1753.7
Pylon-512	521	3452.8

Table 7: Complexities of resultant attack on Pylon ($\omega = 2.8$) [39,34].

Algebraic Attacks Using Linearization Technique. The algebraic attack of Zhang et al. [44] on round-reduced RAIN is based on a guess-and-linearize strategy that exploits the power-map structure of the large inversion S-box. For one-round RAIN, the attacker guesses a suitable value $x^d = \alpha$, where $d \mid (2^n - 1)$, and rewrites the inversion in the form $x^{-1} = (x^d)^s \cdot x^{2^t}$. After guessing α , the term $(x^d)^s$ becomes a constant, while x^{2^t} is an $\mathbb{F}_2$ -linear Frobenius map. As a result, the nonlinear S-box is transformed into a linear expression, and the resulting system can be solved efficiently by Gaussian elimination. For two-round RAIN, the same idea is not applied independently to both S-boxes, since linearizing each inversion layer already incurs a cost of about $(2^n - 1)/d$, which would eliminate the gain over exhaustive search. Instead, the first S-box is linearized to reduce the number of effective variables, and the second S-box is then exploited to derive an overdefined quadratic system in the remaining variables. In this setting, [44] choose divisors of the special form $d = 2^w + 1$, which arise naturally from the factorization of $2^n - 1$ for even n . After guessing $(P + k + c^{(1)})^d = \alpha$, the first S-box becomes linear, and the relation

$$(P + k + c^{(1)})^{2^{2w}} = \alpha^{2^w - 1} (P + k + c^{(1)}) \quad (23)$$

yields additional linear equations. Consequently, the internal state can be expressed using only $2w$ free variables. The second S-box is then represented as quadratic equations in these free variables, and the resulting system is finally solved by standard linearization and Gaussian elimination.

Our parameter choices are specifically intended to rule out the above guess-and-linearize strategy. First, unlike the parameter sets targeted in [44], all our extension degrees n are chosen as odd primes. Hence, the special factorization pattern of $2^n - 1$ used in the RAIN attack for even n does not arise in our setting. In particular, the decomposition underlying the linearization of the inverse map, which relies on divisors of the form $d = 2^w + 1$ derived from the even-degree structure of $\mathbb{F}_{2^n}$, is no longer available. Therefore, the first step of the attack, namely reducing the inversion layer to an $\mathbb{F}_2$ -linear expression by guessing a subgroup value $x^d = \alpha$, cannot be carried out in the same way. Second, for our two Mersenne-type exponents e_1 and e_2 , the parameters are selected so as to avoid any nontrivial divisor $d \mid (2^n - 1)$ satisfying $(2^{e_i} - 1)^{-1} \equiv 2^t \pmod{d}$ for some integer t and for either $i \in \{1, 2\}$. Consequently, the S-boxes cannot be rewritten in the form $x^{e_i} = (x^d)^s \cdot x^{2^t}$ with a nontrivial subgroup guess and an $\mathbb{F}_2$ -linear residual term. Moreover, there is no common nontrivial divisor $d \mid (2^n - 1)$ that would simultaneously enable such a decomposition for both exponents e_1 and e_2 . This rules out the simultaneous linearization mechanism that is essential in the attack on full AIM-III in [44]. Hence, for our parameter sets, the algebraic attack of [44] loses its crucial entry point: the adversary can no longer exploit a low-cost subgroup guess to transform one nonlinear layer into a linear system and then derive a sufficiently overdefined quadratic system from the next layer. Therefore, our instances are resistant to the specific linearization-based subgroup attacks developed in [44].

For completeness, we explicitly verify the subgroup-linearization condition for all recommended parameter sets. For an exponent e , define

$$\mathcal{D}_e(n) = \{d > 1 : d \mid (2^n - 1), \exists t \in \{0, \dots, n-1\} \text{ s.t. } (2^e - 1)^{-1} \equiv 2^t \pmod{d}\}. \quad (24)$$

Here the inverse is taken modulo d . Since our parameters satisfy $\gcd(e, n) = 1$, we have

$$\gcd(2^e - 1, 2^n - 1) = 2^{\gcd(e, n)} - 1 = 1, \quad (25)$$

and hence $2^e - 1$ is invertible modulo every divisor of $2^n - 1$. If a composite divisor d satisfies the above condition, then every prime divisor $p \mid d$ also satisfies $(2^e - 1)^{-1} \equiv 2^t \pmod{p}$. Therefore, it suffices to check the condition over all prime divisors of $2^n - 1$. In our verification, we used the complete prime factorizations of $2^n - 1$ for the recommended values of n . The script first verifies that the product of the listed prime factors is exactly $2^n - 1$, and then checks each prime divisor p and each $t \in \{0, \dots, n-1\}$. No nontrivial prime-divisor hit was found. By the prime-divisor reduction above, this implies $\mathcal{D}_{e_i}(n) = \emptyset$ for both $i = 1, 2$ and for all recommended parameter sets.

We also check the divisor pattern used in the linearization attack on reduced RAIN, namely divisors of the form $2^w + 1$ of $2^n - 1$. For our choices of n , this condition is ruled out structurally. Indeed, if $2^w + 1 \mid 2^n - 1$, then $2^w \equiv -1 \pmod{2^w + 1}$, so the multiplicative order of 2 modulo this divisor is even. On the other hand, since the divisor also divides $2^n - 1$, this order must divide the odd prime n , which is impossible. Thus no divisor of the form $2^w + 1$ occurs for our parameter sets. As shown in Table 8, none of our recommended parameter sets admits a nontrivial subgroup divisor enabling the decomposition $x^{\bar{e}_i} = (x^d)^s \cdot x^{2^t}$ with an $\mathbb{F}_2$ -linear residual Frobenius term. Consequently, the guess-and-linearize step required by the attacks of [44] is not available for Pylon.

Variant	n	(e_1, e_2)	$\exists w : 2^w + 1 \mid 2^n - 1$	$\mathcal{D}_{e_1}(n)$	$\mathcal{D}_{e_2}(n)$
Pylon-128	137	(70, 75)	none	$\emptyset$	$\emptyset$
Pylon-192	197	(94, 105)	none	$\emptyset$	$\emptyset$
Pylon-256	263	(129, 136)	none	$\emptyset$	$\emptyset$
Pylon-512	521	(248, 273)	none	$\emptyset$	$\emptyset$

Table 8: Parameter checks against subgroup-linearization attacks. The fourth column checks if a divisor of the form $2^w + 1$ divides $2^n - 1$. The sets $\mathcal{D}_{e_i}(n)$ contain nontrivial divisors $d \mid 2^n - 1$ satisfying $(2^{e_i} - 1)^{-1} \equiv 2^t \pmod{d}$ for some $t \in \{0, \dots, n-1\}$.

6.3 Statistical Attacks

In the considered one-wayness setting, the adversary is given a target pair $(\text{iv}_{\text{owf}}, ct)$ generated from an unknown secret input pt . The goal is to find some preimage pt' such that $\text{Pylon}(\text{iv}_{\text{owf}}, pt') = ct$. This is different from the usual

block-cipher cryptanalysis setting, where an adversary may obtain many input-output pairs correlated with the same secret key or secret instance. Although Pylon is public and can be evaluated offline on arbitrary inputs, such evaluations do not by themselves provide many samples correlated with the unknown target preimage. Therefore, classical statistical distinguishers would have to be converted into preimage-finding algorithms in order to be useful in our setting. We are not aware of such a conversion that improves over generic preimage search for Pylon.

Differential cryptanalysis. Differential cryptanalysis relies on observing how input differences propagate through many correlated input-output pairs. In our setting, the attacker is given only one target value and aims to recover a preimage. The attacker can of course evaluate the public function on arbitrary inputs, but these evaluations do not give difference pairs involving the unknown preimage pt . Hence, a standard differential attack does not directly translate into a preimage-recovery attack against Pylon in the single-target one-wayness model.

Linear cryptanalysis. Linear cryptanalysis requires estimating biases of linear approximations from many correlated input-output samples. In the one-wayness setting considered here, only one target input-output relation involving the hidden preimage is available. Offline evaluations of the public primitive may reveal global statistical properties of the function, but they do not directly reveal whether the hidden preimage satisfies a useful linear approximation. Therefore, standard linear cryptanalysis does not give a direct advantage over generic preimage search unless such a distinguisher can be turned into an efficient inversion algorithm.

Integral attacks. Integral attacks exploit balancedness or cancellation properties over carefully chosen structured sets of inputs. Such attacks are naturally formulated in settings where the adversary can observe the propagation of an entire structured set through the same secret instance. In our attack model, the adversary only has one target value associated with the unknown preimage. Although the public function can be evaluated on chosen structured sets, this does not directly test whether the hidden preimage belongs to a useful integral structure. Hence, standard integral attacks are not directly applicable in their usual form.

Saturation attacks. Saturation attacks are closely related to integral attacks and similarly rely on tracking how structured sets of inputs propagate through the primitive. Since the one-wayness experiment provides only a single target output and no structured set of correlated evaluations involving the secret preimage, saturation attacks also do not directly yield a preimage-finding method in our model.

Consequently, differential, linear, integral, and saturation attacks do not directly apply in their standard forms to the single-target one-way-function setting considered in this work. In particular, these attacks require many correlated

input-output samples, whereas the adversary in our model is given only a single target value associated with the unknown preimage. Offline evaluations of the public function do not provide such correlated samples involving the hidden preimage, and hence do not yield an effective preimage recovery strategy.

6.4 Quantum Attacks

There are two types of quantum attacks. In the first case, an adversary is allowed to use quantum computation but not allowed to make any quantum query, while in the second case both are accessible [43].

Quantum Algebraic Attack. Since each inverse Mersenne S-box contributes $3n$ linearly independent quadratic equations like the S-box in AIM [26] and AIM2 [27], quantum algebraic attacks such as GröverXL algorithm [11] and Quantum-BooleanSolve algorithm [19] yield a complexity of approximately $O(2^{(0.3687+o(1)) \cdot 3n})$ and $O(2^{0.462 \cdot 3n})$ respectively, which is significantly higher than the generic Grover search complexity of $O(2^{0.5n})$ [24]. Hence we conclude that the most effective quantum attack remains the generic Grover search [24].

Quantum Generic Attack. Advanced quantum generic attacks, particularly those based on Simon’s period-finding algorithm [35] or its offline variants [12], rely on querying a keyed oracle multiple times to detect hidden periods. However, in our setting, the adversary is restricted to a single public target pair (iv_{owf}, ct) . As the adversary lacks access to a keyed oracle for performing multiple classical or superposition queries, constructing the periodic function required for Simon’s algorithm [35] is impossible. Therefore, structural quantum attacks are inapplicable, and the generic security of our primitive is still determined by Grover’s algorithm [24], providing $n/2$ bits of security.

6.5 Summary of Dedicated Attack Analysis

Table 9 summarizes the dedicated attacks considered above in the single-target one-way-function setting. Attacks that inherently require many input-output pairs are marked as not applicable.

This table shows that none of the dedicated attacks considered above improves upon the claimed classical security level for the recommended parameter sets. The next subsection complements this concrete analysis with a generic one-wayness proof in the public permutation model.

6.6 Formal Security Analysis

6.6.1 Security Models and Scope of the Generic proof

Before stating the formal proofs, we make explicit the scope of the proofs in this part and the relation to the concrete cryptanalysis in Sect. 6.

Attack type	Estimate / status	Conclusion
Brute-force preimage search	2^n	Generic classical baseline
Gröbner basis over $\mathbb{F}_2$	Table 4	Above target
Gröbner basis over $\mathbb{F}_{2^n}$	$> 2^n$	Not competitive
Overdefined quadratic modeling	No RAIN-type overdefinition	Ineffective
MITM resultant modeling	Degree blow-up	Not competitive
Direct resultant attack	Table 7	Above target
Subgroup linearization	Table 8	Ruled out by parameters
Differential / linear cryptanalysis	N/A	Require many correlated samples
Integral / saturation attacks	N/A	Require structured input sets
Quantum algebraic attacks	Worse than Grover	Not best known
Generic quantum preimage search	$2^{n/2}$	Quantum baseline

Table 9: Summary of dedicated attacks on Pylon. “N/A” means that the attack does not apply in the single-target one-way-function setting.

Two complementary analyses. The security of Pylon is supported by two independent lines of argument:

- *Structural cryptanalysis (Sect. 6).* We present algebraic attacks (Gröbner basis, overdefined quadratic modeling, resultant attacks, linearization, statistical attacks etc.) that exploit the specific structure of the inverse Mersenne S-boxes, the dense linear layers, and the nested feed-forward connections. These attacks treat Pylon as the concrete function it is.
- *Generic black-box analysis (This section).* We idealize S_1, S_2 as independent uniformly random permutations and bound the success probability of any PPT adversary that interacts with them only as oracles. This rules out generic black-box preimage strategies, but says nothing about attacks exploiting the algebraic structure of the actual nonlinear operations.

Why both are needed? The idealized-permutation analysis is necessary but not sufficient. It is necessary because even an attacker who does not exploit the algebraic structure must do with at least $\mathcal{O}(2^n/q)$ complexity per query on average. It is not sufficient because the S-box layer is a fixed algebraic permutation, not a random one. Any complete security claim must combine both: the dedicated cryptanalysis of Sect. 6 establishes that no known structural attack improves over the generic bound for the recommended parameters. We emphasize this scope distinction upfront because the result of Theorem 1 below is called “generic one-wayness.” This terminology is standard but can mislead: it refers to security against attackers that use only black-box access, not to unconditional one-wayness.

6.6.2 Generic Security Bounds of Pylon under the Ideal-Permutation Model

Model. We model S_1 and S_2 as independent, uniformly random permutations over $\{0, 1\}^n$, evaluated by a *lazy sampling* challenger: each oracle query to S_ℓ (or S_ℓ^{-1}) at a fresh input is answered by drawing the response uniformly and

independently from the set of output values not yet assigned by S_ℓ . Without loss of generality, the adversary $\mathcal{A}$ is deterministic (fixing the randomness that maximizes its advantage), makes exactly q queries (padding with fresh dummy queries if necessary), and never repeats a query or queries S_ℓ and S_ℓ^{-1} on values forced to coincide by prior responses. We make two further conventions. (i) *Augmented adversary*. We replace $\mathcal{A}$ by an augmented adversary $\mathcal{A}^+$ that, just before outputting $\mathbf{pt}'$, additionally queries the complete S_1 - and S_2 -path of $\mathbf{pt}'$. This adds at most two queries and does not decrease the success probability, so we may assume any winning $\mathbf{pt}'$ induces a consistent pair in $\mathcal{A}^+$'s transcript $(\mathcal{T}_1, \mathcal{T}_2)$; we write $Q \leq q + 2$ for the resulting number of queries. (ii) *Challenge path*. When forming $\mathbf{ct} = \text{Pylon}(\text{iv}_{\text{owf}}, \mathbf{pt})$, the challenger internally fixes the two oracle values on the challenge path, $(a^* = M_1(\mathbf{pt}) \oplus \gamma_1, S_1(a^*))$ and $(c^*, S_2(c^*))$. These planted assignments are *not* part of the adversary-visible transcript $(\mathcal{T}_1, \mathcal{T}_2)$; they enter it only if the adversary queries the corresponding points. Let HitPath be the event that some query of $\mathcal{A}^+$ lands on a challenge-path input or output. Lazy sampling is kept consistent with the planted assignments only on HitPath .

Game 1 (Preimage-finding of Pylon). *The experiment $\text{Exp}_{\text{Pylon}}^{\text{pre}}(\mathcal{A})$ proceeds as follows.*

1. *The challenger samples $S_1, S_2 \xleftarrow{\$} \text{Perm}(\{0, 1\}^n)$ independently, $\text{iv}_{\text{owf}} \xleftarrow{\$} \{0, 1\}^n$, and $\mathbf{pt} \xleftarrow{\$} \{0, 1\}^n$, then computes $\mathbf{ct} := \text{Pylon}(\text{iv}_{\text{owf}}, \mathbf{pt})$ and sends $(\text{iv}_{\text{owf}}, \mathbf{ct})$ to $\mathcal{A}$ (matching KeyGen , where iv_{owf} is part of $\mathbf{pk}$).*
2. *$\mathcal{A}$ makes adaptive queries to $S_1, S_1^{-1}, S_2, S_2^{-1}$.*
3. *$\mathcal{A}$ outputs $\mathbf{pt}'$.*

$\mathcal{A}$ wins if $\text{Pylon}(\text{iv}_{\text{owf}}, \mathbf{pt}') = \mathbf{ct}$. The advantage is $\mathbf{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{A}) := \Pr[\mathcal{A} \text{ wins}]$.

Let $\mathcal{T}_1 = \{(a_i, y_i)\}$ and $\mathcal{T}_2 = \{(c_j, z_j)\}$ denote the input-output records of all $\mathcal{A}$'s S_1/S_1^{-1} -queries and S_2/S_2^{-1} -queries respectively, with $|\mathcal{T}_1| + |\mathcal{T}_2| = q$. Each entry satisfies $S_1(a_i) = y_i$ (resp. $S_2(c_j) = z_j$).

Consistent pairs. We say $(\mathcal{T}_1, \mathcal{T}_2)$ contains a consistent pair for $\mathbf{ct}$ if there exist $(a, y) \in \mathcal{T}_1$ and $(c, z) \in \mathcal{T}_2$ satisfying

$$c = M_2(y \oplus M_1^{-1}(a \oplus \gamma_1)) \oplus \text{iv}_{\text{owf}}, \quad (26)$$

$$M_3(z) \oplus c \oplus M_1^{-1}(a \oplus \gamma_1) \oplus \gamma_2 = \mathbf{ct}. \quad (27)$$

Given such a pair, the preimage candidate $\mathbf{pt}^* := M_1^{-1}(a \oplus \gamma_1)$ satisfies $\text{Pylon}(\text{iv}_{\text{owf}}, \mathbf{pt}^*) = \mathbf{ct}$. Because $\mathcal{A}$ can only evaluate S_1 and S_2 through its oracle queries, any output $\mathbf{pt}'$ with $\text{Pylon}(\text{iv}_{\text{owf}}, \mathbf{pt}') = \mathbf{ct}$ requires

$$(M_1(\mathbf{pt}') \oplus \gamma_1, S_1(M_1(\mathbf{pt}') \oplus \gamma_1)) \in \mathcal{T}_1 \quad \text{and} \quad (M_2(\cdot) \oplus \text{iv}_{\text{owf}}, S_2(M_2(\cdot) \oplus \text{iv}_{\text{owf}})) \in \mathcal{T}_2.$$

Hence $\mathbf{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{A}) \leq \Pr[(\mathcal{T}_1, \mathcal{T}_2) \text{ contains a consistent pair for } \mathbf{ct}]$.

Theorem 1 (Generic black-box (linear-query) preimage bound for Pylon under the ideal-permutation model). *If S_1, S_2 are independent uniformly random permutations of $\{0, 1\}^n$, then for any (computationally unbounded) adversary $\mathcal{A}$ making at most q oracle queries in total, with $q < 2^{n-1}$,*

$$\text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{A}) = O\left(\frac{q}{2^n}\right), \quad \text{more precisely} \quad \text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{A}) \leq \frac{3(q+2)}{2^n - q - 2}.$$

Proof. We bound $\Pr[(\mathcal{T}_1, \mathcal{T}_2) \text{ contains a consistent pair for ct}]$ by a *last-query charging* argument. The crucial structural fact is that a consistent pair is fixed by a *single, joint* match, not by two independent single-coordinate collisions.

Unique completing partner. Given a record $(a, y) \in \mathcal{T}_1$, put $\text{pt} = M_1^{-1}(a \oplus \gamma_1)$. Then (26) fixes a unique $c = M_2(y \oplus \text{pt}) \oplus \text{iv}_{\text{owf}}$, and (27) fixes a unique $z = M_3^{-1}(\text{ct} \oplus c \oplus \text{pt} \oplus \gamma_2)$; hence each $\mathcal{T}_1$ -record has at most one completing $\mathcal{T}_2$ -record. Symmetrically, given $(c, z) \in \mathcal{T}_2$, the values

$$\text{pt} = \text{ct} \oplus M_3(z) \oplus c \oplus \gamma_2, \quad a = M_1(\text{pt}) \oplus \gamma_1, \quad y = M_2^{-1}(c \oplus \text{iv}_{\text{owf}}) \oplus \text{pt}$$

determine a unique completing $\mathcal{T}_1$ -record. Thus a consistent pair requires the *joint* match of $(a, y) \leftrightarrow (c, z)$ (equivalently of the pair (pt, c)), spanning $2n$ bits; in particular the feed-forward prevents any birthday/meet-in-the-middle split, so no single-coordinate collision suffices.

Charging. For $i = 1, \dots, q$ let CP_i be the event that query i creates a consistent pair absent after queries $1, \dots, i-1$. These events are pairwise disjoint and their union is the consistent-pair event, so $\Pr[\exists \text{ consistent pair}] = \sum_i \Pr[\text{CP}_i]$. We charge each pair to query i , the *later* of its two queries, whose fresh response is uniform over a set of size $\geq 2^n - q$.

S_1 -queries. Take a forward query $S_1(a_i) = y_i$ (the inverse query is identical after swapping the roles of input and response). By uniqueness, query i completes an earlier record $(c, z) \in \mathcal{T}_2^{(i)}$ only if the input matches, $a_i = M_1(\text{pt}(c, z)) \oplus \gamma_1$, and the fresh response equals the value it forces, $y_i = M_2^{-1}(c \oplus \text{iv}_{\text{owf}}) \oplus \text{pt}(c, z)$. The latter holds with probability at most $1/(2^n - q)$ for each such record, so

$$\Pr[\text{CP}_i \mid i \text{ is an } S_1\text{-query}] \leq \frac{\kappa_i}{2^n - q}, \quad \kappa_i := |\{(c, z) \in \mathcal{T}_2^{(i)} : a(c, z) = a_i\}|.$$

Each $\mathcal{T}_2$ -record determines a single completing input $a(c, z)$, and the S_1 -query inputs are distinct, so $\sum_{i: S_1} \kappa_i \leq |\mathcal{T}_2| = q_2$, giving $\sum_{i: S_1} \Pr[\text{CP}_i] \leq q_2/(2^n - q)$.

S_2 -queries. Symmetrically, $\sum_{i: S_2} \Pr[\text{CP}_i] \leq q_1/(2^n - q)$.

Combining. With $q_1 + q_2 = q$,

$$\sum_{i=1}^q \Pr[\text{CP}_i] \leq \frac{q_2}{2^n - q} + \frac{q_1}{2^n - q} = \frac{q}{2^n - q}.$$

Conditioned on $\neg \text{HitPath}$, every response of $\mathcal{A}^+$ is fresh and independent of the planted challenge path, so the charging above applies with q replaced by

Q , giving $\Pr[\exists \text{ consistent pair} \mid \neg \text{HitPath}] \leq Q/(2^n - Q)$. Each challenge-path point is a uniform value hidden from $\mathcal{A}^+$, so $\Pr[\text{HitPath}] \leq 2Q/(2^n - Q)$. Since a winning pt' induces a consistent pair in $\mathcal{A}^+$'s transcript,

$$\text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{A}) \leq \Pr[\text{HitPath}] + \Pr[\exists \text{ consistent pair} \mid \neg \text{HitPath}] \leq \frac{3Q}{2^n - Q} \leq \frac{3(q+2)}{2^n - q - 2}.$$

The bound is *linear* in q : a generic preimage requires $q = \Theta(2^n)$ queries, i.e. about n bits of generic classical preimage security. (The earlier $O(q^2/2^n)$ estimate over-charged each fresh response against all q_2 opposite-side records; since the completing partner is unique, the correct charge is $O(q/2^n)$.) $\square$

Remark 1 (Multi-user setting). Consider $u \geq 1$ users sharing the same permutations S_1, S_2 , linear layers M_1, M_2, M_3 , and constants γ_1, γ_2 , but with independently sampled key pairs $\text{iv}_k \xleftarrow{\$} \{0, 1\}^n$, $\text{pt}_k \xleftarrow{\$} \{0, 1\}^n$ and targets $\text{ct}_k = \text{Pylon}(\text{iv}_k, \text{pt}_k)$; the adversary wins if it inverts *any* one target. The single-target analysis applies per target (replacing $(\text{iv}_{\text{owf}}, \text{ct})$ by $(\text{iv}_k, \text{ct}_k)$ changes only constant offsets in the completing-partner maps). A union bound over the u targets gives

$$\text{Adv}_{\text{Pylon}}^{\text{pre}, u}(\mathcal{A}) \leq u \cdot \frac{O(q)}{2^n - q} = O\left(\frac{uq}{2^n}\right).$$

6.6.3 EUF-CMA Security of GreatWall in the Random Oracle Model.

We prove the EUF-CMA security of GreatWall in the classical random oracle model (ROM). Conditional on the stated properties of the underlying VOLEitH proof system (Definition 2), the reduction reduces EUF-CMA security to the preimage resistance of Pylon.

Background. Let $R_{\text{Pylon}} := \{(pt, (\text{iv}_{\text{owf}}, ct)) : \text{Pylon}(\text{iv}_{\text{owf}}, pt) = ct\}$ be the relation witnessed by Pylon. A GreatWall signature is a non-interactive proof obtained via the Fiat–Shamir transform [20] applied to the VOLEitH interactive proof of knowledge for R_{Pylon} . The public key of GreatWall is $(\text{iv}_{\text{owf}}, ct)$ and the secret key is the corresponding preimage pt . The hash functions are modeled as classical random oracles.

Definition 2 (Required properties of the VOLEitH proof system over R_{Pylon} , classical ROM). We require the following standard properties of the VOLEitH proof system instantiated for R_{Pylon} with parameters $(\tau, N, T_{\text{open}}, w_{\text{grind}})$:

1. **Knowledge soundness with (online) extraction error ε_{ks} .** For any PPT adversary $\mathcal{A}^*$ that produces an accepting Fiat–Shamir proof π for public statement $pk = (\text{iv}_{\text{owf}}, ct)$ with non-negligible probability δ , there exists a PPT online (straight-line) knowledge extractor Ext that, given the list of $\mathcal{A}^*$'s random-oracle queries together with a single accepting proof π , outputs a witness pt' with $\text{Pylon}(\text{iv}_{\text{owf}}, pt') = ct$ with probability at least $\delta - \varepsilon_{ks}$.

(For the VOLE-in-the-head proof system, knowledge soundness is established via straight-line extraction from the committed GGM seeds recorded in the random-oracle table, not via rewinding/forking.)

The extraction error ε_{ks} is precisely the Fiat-Shamir soundness error of the VOLEitH proof system instantiated with parameters $(\tau, N, T_{\text{open}}, w_{\text{grind}})$ and a VOLE consistency hash of bit length $\lambda + B$; its exact expression, as a function of these parameters and the number q_H of random-oracle queries, is given by the FAEST/VOLEitH-with-one-tree soundness analysis [10, 7]. The soundness is established by straight-line (online) extraction, and the parameters w_{grind} and T_{open} are tuned to control this error; for the parameter sets in Table 2, that analysis yields $\varepsilon_{ks} \leq 2^{-\lambda}$.

2. **Honest-verifier zero-knowledge (HVZK) simulation.** There exists a PPT simulator Sim that, given only $pk = (iv_{\text{owf}}, ct)$ and a message M , produces a signature transcript $\tilde{\sigma}$ together with a list of random-oracle points to program. The joint distribution of $\tilde{\sigma}$ and the (programmed) random oracle is ε_{sim} -indistinguishable from a real signature, with

$$\varepsilon_{\text{sim}} \leq \frac{q_S \cdot q_H}{2^\lambda} + \frac{q_S^2}{2^\lambda},$$

where q_S is the number of signature queries. The first term bounds the probability that $\mathcal{A}$ queries the random oracle at a programmed point before the signing oracle programs it; the second bounds collisions among programmed points across signature queries.

3. **Preimage resistance of Pylon.** For every PPT adversary $\mathcal{B}$:

$$\text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{B}) := \Pr [\text{Pylon}(iv_{\text{owf}}, pt') = ct],$$

where $pt \xleftarrow{\$} \{0, 1\}^n$, $iv_{\text{owf}} \xleftarrow{\$} \{0, 1\}^n$, $ct := \text{Pylon}(iv_{\text{owf}}, pt)$, and $pt' \leftarrow \mathcal{B}(iv_{\text{owf}}, ct)$. By Theorem 1, a $\mathcal{B}$ making q_B ideal-permutation queries satisfies $\text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{B}) = O(q_B/2^n)$; together with the dedicated cryptanalysis of Sect. 6, which shows that no structural attack improves on this generic bound, the constant-success query complexity is $\Theta(2^n)$. Since $n \geq \lambda$ for all recommended instances, the generic classical preimage cost is at least 2^λ .

EUFCMA experiment. The experiment $\text{Exp}_{\text{GreatWall}}^{\text{EUFCMA}}(\mathcal{A})$ gives the PPT adversary $\mathcal{A}$ access to a signing oracle $\mathcal{O}^{\text{Sign}}(\cdot)$ that signs adaptively chosen messages under the honestly generated key pair (pk^*, sk^*) , where $pk^* = (iv_{\text{owf}}^*, ct^*)$ and $sk^* = pt^*$. The adversary also has access to all random oracles. The adversary wins if it outputs (M^*, σ^*) such that $M^* \notin \mathcal{Q}$, where $\mathcal{Q}$ is the set of queried messages, and $\text{Verify}(pk^*, M^*, \sigma^*) = 1$. The advantage is $\text{Adv}_{\text{GreatWall}}^{\text{EUFCMA}}(\mathcal{A}) := \Pr[\mathcal{A} \text{ wins}]$.

Theorem 2 (Reduction of EUFCMA security to the preimage resistance of Pylon). Under the conditions of Definition 2, for any PPT adversary $\mathcal{A}$ making at most q_S signature queries and q_H random-oracle queries, there exists a PPT adversary $\mathcal{B}$ against the preimage resistance of Pylon, with running time at most $T(\mathcal{B}) \leq T(\mathcal{A}) + (q_S + q_H) \cdot \text{poly}(\lambda)$, such that

$$\text{Adv}_{\text{GreatWall}}^{\text{EUFCMA}}(\mathcal{A}) \leq \text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{B}) + \varepsilon_{ks} + \varepsilon_{\text{sim}} + \frac{q_S q_H}{2^\lambda}.$$

Proof. We proceed via a sequence of games.

Game 2. The original $\text{Exp}_{\text{GreatWall}}^{\text{EUF-CMA}}(\mathcal{A})$ experiment. The key pair (pk^*, sk^*) is generated honestly, where $pk^* = (iv_{\text{owf}}^*, ct^*)$ and $sk^* = pt^*$, and the signing oracle uses sk^* . By definition:

$$\Pr[\mathcal{A} \text{ wins in Game 2}] = \text{Adv}_{\text{GreatWall}}^{\text{EUF-CMA}}(\mathcal{A}).$$

Game 3. Identical to Game 2, except that all signature queries are answered by the simulator Sim of Definition 2, which does not use pt^* . The simulator programs random-oracle queries as needed. By the HVZK guarantee:

$$|\Pr[\mathcal{A} \text{ wins in Game 3}] - \Pr[\mathcal{A} \text{ wins in Game 2}]| \leq \varepsilon_{\text{sim}}.$$

Game 4. Identical to Game 3, except that the game aborts if the λ -bit Fiat–Shamir challenge associated with the forgery coincides with one of the (at most q_S) challenges generated during the signing queries. Each signing challenge is a uniform λ -bit string and the forgery challenge is fixed by one of $\mathcal{A}$'s at most q_H random-oracle queries; a union bound over all (random-oracle query, signing query) pairs gives:

$$|\Pr[\mathcal{A} \text{ wins in Game 4}] - \Pr[\mathcal{A} \text{ wins in Game 3}]| \leq \frac{q_S q_H}{2^\lambda}.$$

Reduction. We now construct $\mathcal{B}$ from $\mathcal{A}$'s success in Game 4; see Algorithm 20.

Correctness of the simulation. $\mathcal{B}$ provides $\mathcal{A}$ with exactly the view of Game 4: the public key pk^* is the challenge from the preimage-resistance experiment, signatures are produced by Sim with consistent oracle programming, and the collision-check abort is enforced as in Game 4.

Analysis of success. If $\mathcal{A}$ wins in Game 4, then σ^* is an accepting Fiat–Shamir transcript for (pk^*, M^*) . By Definition 2, the extractor Ext outputs a valid witness pt' with $\text{Pylon}(iv_{\text{owf}}^*, pt') = ct^*$ except with probability ε_{ks} . Hence:

$$\Pr[\mathcal{B} \text{ wins}] \geq \Pr[\mathcal{A} \text{ wins in Game 4}] - \varepsilon_{ks}.$$

Running time. $\mathcal{B}$'s overhead consists of:

- Maintaining the random-oracle table ($O(q_H \cdot \text{poly}(\lambda))$);
- Running Sim for each of the q_S signature queries ($O(q_S \cdot \text{poly}(\lambda))$);
- Running the online extractor Ext , which post-processes the recorded random-oracle table (no rewinding).

Since the online extractor of Definition 2 runs $\mathcal{A}$ only once and then post-processes the random-oracle table, $T(\mathcal{B}) \leq T(\mathcal{A}) + (q_S + q_H) \cdot \text{poly}(\lambda)$.

Algorithm 20: Preimage-finding adversary $\mathcal{B}(pk^*)$ in the ROM

Input: Pylon preimage challenge $pk^* = (iv_{\text{owf}}^*, ct^*)$; oracle access to $\mathcal{A}$.
Output: A candidate preimage pt' such that $\text{Pylon}(iv_{\text{owf}}^*, pt') = ct^*$, or $\perp$.

```

1  $\mathcal{Q} \leftarrow \emptyset$ ;
2 Initialize an empty internal table for the random oracles;
3 Give  $pk^*$  to  $\mathcal{A}$  as the public key;
4 /* Random-oracle queries */
5 foreach random-oracle query made by  $\mathcal{A}$  to some  $H$  at point  $x$  do
6   if  $x$  is not yet in the internal table then
7     | Sample a uniformly random response, store it, and return it to  $\mathcal{A}$ ;
8   else
9     | Return the previously stored value to  $\mathcal{A}$ ;

10 /* Signing queries */
11 foreach classical signing query  $M$  from  $\mathcal{A}$  do
12    $\sigma \leftarrow \text{Sim}(pk^*, M)$ ; // Sim returns simulated  $\sigma$  plus a list of RO
    points to program
13   Program the internal random-oracle table at those points to the values
    specified by Sim;
14   if programming would overwrite an existing distinct entry then
15     | return  $\perp$ ; // captured by  $\varepsilon_{\text{sim}}$ 
16    $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{M\}$ ;
17   Return  $\sigma$  to  $\mathcal{A}$ ;

18 /* Forgery phase */
19 Receive  $(M^*, \sigma^*)$  from  $\mathcal{A}$ ;
20 if  $M^* \in \mathcal{Q}$  or  $\text{Verify}(pk^*, M^*, \sigma^*) \neq 1$  or the collision-check abort fires then
21   | return  $\perp$ ;
22 Run the online knowledge extractor  $\text{Ext}$  from Definition 2(1) on the recorded
    random-oracle table and the accepting transcript  $(pk^*, M^*, \sigma^*)$ ;
23 Obtain a candidate witness  $pt'$ ;
24 return  $pt'$ ;
```

Combining. From the game hops and the reduction:

$$\begin{aligned}
\text{Adv}_{\text{GreatWall}}^{\text{EUF-CMA}}(\mathcal{A}) &= \Pr[\mathcal{A} \text{ wins in Game 2}] \\
&\leq \Pr[\mathcal{A} \text{ wins in Game 3}] + \varepsilon_{\text{sim}} \\
&\leq \Pr[\mathcal{A} \text{ wins in Game 4}] + \varepsilon_{\text{sim}} + \frac{q_S q_H}{2^\lambda} \\
&\leq \Pr[\mathcal{B} \text{ wins}] + \varepsilon_{ks} + \varepsilon_{\text{sim}} + \frac{q_S q_H}{2^\lambda} \\
&= \text{Adv}_{\text{Pylon}}^{\text{pre}}(\mathcal{B}) + \varepsilon_{ks} + \varepsilon_{\text{sim}} + \frac{q_S q_H}{2^\lambda}. \quad \square
\end{aligned}$$

Remark 2 (Concrete parameters). For the recommended parameter sets in Table 2, the reduction gives

$$Adv_{\text{GreatWall}}^{\text{EUF-CMA}}(\mathcal{A}) \leq Adv_{\text{Pylon}}^{\text{pre}}(\mathcal{B}) + \varepsilon_{\text{ks}} + \varepsilon_{\text{sim}} + \frac{q_S q_H}{2^\lambda}.$$

The parameters w_{grind} , T_{open} , and the $(\lambda + B)$ -bit VOLE consistency hash are chosen so that ε_{ks} is negligible for the considered query budget. The concrete security claim therefore follows from this explicit bound together with the preimage resistance of Pylon analyzed in Section 6.

6.6.4 Post-Quantum Extension: EUF-CMA Security in the QROM

Since GreatWall is designed as a post-quantum signature scheme, the EUF-CMA security claim must also hold against quantum adversaries with quantum access to the random oracles. We now state the corresponding quantum random oracle (QROM) security claim, which follows by replacing the classical extractor and simulator of Definition 2 with their QROM counterparts established in recent work on the post-quantum Fiat–Shamir transform [18,30,23,42].

Remark. The QROM analysis follows the standard Fiat–Shamir framework for VOLEitH-based signatures. In particular, we rely on the QROM security analysis of Fiat–Shamir transformations established in [18,30,23,42] and on the security framework of VOLEitH-based signatures [9,7]. The only additional ingredient specific to GreatWall is the reduction from a successful forgery to a preimage of the underlying Pylon one-way function.

Why a separate post-quantum analysis is needed? Although our classical reduction (Theorem 2) already uses *online* (straight-line) extraction, it does so by reading the *classical* list of random-oracle queries; under quantum access the adversary queries the random oracles in superposition, so this query list is no longer well-defined and both simulation and extraction must be re-established with quantum-accessible oracles. Moreover, the older Fiat–Shamir analyses based on rewinding/forking are generally believed to be problematic in the QROM [1], and a generic “lift” from ROM to QROM does *not* apply to Fiat–Shamir signatures without further analysis. Recent advances [18,30,23] establish that VOLEitH-style proof systems admit *online extractability* in the QROM via Zhandry’s compressed-oracle technique, yielding a straight-line reduction. This is the technical foundation for the post-quantum security of all major VOLEitH-based signature schemes, including FAEST [10], FAESTER/MandaRain [7], and AIMer [27]; GreatWall inherits the same framework.

Definition 3 (QROM-analogous properties of the VOLEitH proof system). *We require the QROM analogues of Definition 2:*

1. **Online extractability in the QROM** [18,23]. *There exists a polynomial-time online extractor Ext^{QROM} that, given the compressed-oracle state recorded*

throughout an adversary's quantum interaction with the random oracles together with a single accepting Fiat–Shamir proof π for ct , outputs a witness pt' such that $\text{Pylon}(iv_{owf}, pt') = ct$ except with probability ε_{ks}^Q . The QROM extraction error ε_{ks}^Q is the quantum Fiat–Shamir soundness error of the same VOLEith-with-one-tree proof system, established via QROM online extractability [18, 23] applied to [10, 7]. It coincides with the classical ε_{ks} of Definition 2 up to an additive compressed-oracle term of order $q_H^2/2^\lambda$; for the parameter sets in Table 2, ε_{ks}^Q is negligible for the considered quantum query budget.

2. **HVZK with online simulation in the QROM** [30, 18]. There exists a polynomial-time simulator Sim^{QROM} that, given only $pk = (iv_{owf}, ct)$, produces signature transcripts together with a compressed-oracle reprogramming strategy, such that the joint distribution is ε_{sim}^Q -indistinguishable from a real signing transcript, with

$$\varepsilon_{sim}^Q \leq \frac{q_S \cdot q_H}{2^\lambda} + \frac{q_S^2}{2^\lambda}.$$

3. **Quantum preimage resistance of Pylon.** For every quantum polynomial-time adversary $\mathcal{B}$ making at most q_O quantum evaluations of Pylon, $\text{Adv}_{\text{Pylon}}^{\text{pre}, Q}(\mathcal{B}) = O(q_O^2/2^n)$; equivalently, the constant-success quantum query complexity is $\Theta(2^{n/2})$, matching Grover search. For the recommended parameters this yields the quantum preimage-security levels of Table 3 ($\approx n/2$ bits); see Sect. 6.4 for structural quantum attacks.

Theorem 3 (EUF-CMA security of GreatWall in the QROM). Under the conditions of Definition 3, for any quantum polynomial-time adversary $\mathcal{A}$ making at most q_S classical signature queries and q_H quantum random-oracle queries, there exists a quantum polynomial-time adversary $\mathcal{B}$ against the quantum preimage resistance of Pylon, with running time at most $T(\mathcal{B}) \leq T(\mathcal{A}) + (q_S + q_H) \cdot \text{poly}(\lambda)$, such that

$$\text{Adv}_{\text{GreatWall}}^{\text{EUF-CMA}, Q}(\mathcal{A}) \leq \text{Adv}_{\text{Pylon}}^{\text{pre}, Q}(\mathcal{B}) + \varepsilon_{ks}^Q + \varepsilon_{sim}^Q + \frac{q_S q_H}{2^\lambda}.$$

Proof sketch. The proof follows the same game-hop structure as Theorem 2, with two key differences:

- In Game 3, the classical simulator Sim is replaced by the QROM simulator Sim^{QROM} of Definition 3, which uses Zhandry's compressed-oracle technique [42] to reprogram quantum-accessible random oracles consistently with the simulated transcripts.
- In the final reduction, the classical online extractor Ext is replaced by its QROM counterpart Ext^{QROM} of Definition 3, which extracts from the compressed-oracle state instead of the classical query list. As in the classical case, the reduction is straight-line and runs $\mathcal{A}$ only once, giving $T(\mathcal{B}) \leq T(\mathcal{A}) + (q_S + q_H) \cdot \text{poly}(\lambda)$.

The remaining game-hop analysis (collision check on Fiat–Shamir challenges, contribution of ε_{sim}^Q and ε_{ks}^Q) is identical to the classical case. The detailed QROM analysis of the same proof-system components has been carried out for FAEST [10] and FAESTER/MandaRain [7]; conditional on the QROM online-extractability and simulation properties of this exact VOLEitH instantiation (Definition 3), the same analysis carries over to GreatWall, since the proof system depends on the underlying OWF only through the relation specification. $\square$

Remark 3 (Concrete parameters). For the recommended parameter sets in Table 2, the reduction gives

$$\text{Adv}_{\text{GreatWall}}^{\text{EUF-CMA},Q}(\mathcal{A}) \leq \text{Adv}_{\text{Pylon}}^{\text{pre},Q}(\mathcal{B}) + \varepsilon_{ks}^Q + \varepsilon_{sim}^Q + \frac{q_S q_H}{2^\lambda}.$$

The parameters w_{grind} , T_{open} , and the $(\lambda + B)$ -bit VOLE consistency hash are chosen so that ε_{ks}^Q is negligible for the considered query budget. The concrete security claim therefore follows from this explicit bound together with the preimage resistance of Pylon analyzed in Sect. 6.

Remark 4 (Modularity of the post-quantum claim). The QROM analysis of GreatWall is fully modular: the QROM properties in Definition 3 concern only the VOLEitH proof system, not the Pylon OWF. As such, GreatWall’s post-quantum security inherits directly from the QROM analyses of the underlying VOLEitH framework [10,7], and benefits automatically from any future improvement to those analyses (e.g., tighter QROM bounds or stronger simulator security). The only Pylon-specific post-quantum claim required is the quantum preimage resistance of Definition 3, which is established in Sect. 6.4 through analysis of Grover-style and algebraic quantum attacks.

7 Implementation and Performance

7.1 The Benchmarking Environments

The benchmarks were conducted on the same computing environment of Ubuntu 22.04.5 LTS. The platform is equipped with an Intel Core i9-14900HX @2.42 GHz and 15GB of RAM. All implementations were compiled with GCC 11.4.0, optimized implementations compiled with the AVX2 instructions. Our implementation is mainly referring to faest-one-tree repository¹

7.2 Performance of GreatWall

As noted in the previous sections, the serial iterative structure of Pylon allows it to achieve its full potential when operating over large finite fields. In addition, the structure scales naturally to larger fields and no changes are required except for adjusting the parameters. The performance of Optimized GreatWall

¹ <https://github.com/faest-sign/faest-one-tree>

is summarized in Table 10 and the Reference version is in Table 11. The results show that the proposed construction achieves a noticeable reduction in signature size under larger-field instantiations, while maintaining practical signing and verification performance.

Scheme	$(\tau, w, T_{\text{open}})$	pk (B)	KeyGen (ms)	Sign (ms)	Verify (ms)	Size (B)
GreatWall-128S	(11, 7, 100)	36	0.59	4.59	6.87	2758
GreatWall-128F	(16, 8, 108)	36	0.54	1.04	1.26	3396
GreatWall-192S	(16, 8, 183)	50	1.34	12.08	18.11	6804
GreatWall-192F	(24, 8, 184)	50	1.34	2.57	2.62	8012
GreatWall-256S	(22, 6, 246)	66	1.86	14.57	23.10	12236
GreatWall-256F	(32, 7, 248)	66	1.88	4.26	4.13	14260
GreatWall-512S	(44, 0, 512)	132	16.79	392.83	478.59	50012
GreatWall-512F	(64, 0, 512)	132	17.06	80.90	74.26	57812

Table 10: Performance of Optimized GreatWall for different parameter sets. The suffixes S and F denote slow and fast variants, respectively: S is optimized for shorter signatures, while F is optimized for faster signing and verification.

Scheme	$(\tau, w, T_{\text{open}})$	pk (B)	KeyGen (ms)	Sign (ms)	Verify (ms)	Size (B)
GreatWall-128S	(11, 7, 100)	36	1.268	111.114	89.599	2758
GreatWall-128F	(16, 8, 108)	36	1.215	27.973	19.568	3396
GreatWall-192S	(16, 8, 183)	50	3.112	304.655	266.985	6804
GreatWall-192F	(24, 8, 184)	50	3.147	90.280	57.219	8012
GreatWall-256S	(22, 6, 246)	66	5.013	565.605	480.228	12236
GreatWall-256F	(32, 7, 248)	66	5.045	208.265	129.961	14260
GreatWall-512S	(44, 0, 512)	132	19.993	6087.004	5156.342	50012
GreatWall-512F	(64, 0, 512)	132	19.786	2424.320	1466.726	57812

Table 11: Performance of the Reference implementation for different parameter sets of GreatWall .

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